Pharmaceutical Calculations Booklet

Lectures 1 – 8

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Each lecture is split into topics. Every topic has **Lecture Problems**, then **Textbook Problems**. Answers are at the end of each lecture.

1 Introduction, Units & Equivalences

In this lecture

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1.1 Proportional Sets & Ratios

Lecture Problems

Slide 3 A drug has a concentration of 100 mg/5 ml. Complete the table:

Weight of drug (mg)	Volume (ml)
20	?
50	?
75	?
150	?

a. Volume that contains 20 mg = ?

b. Volume that contains 50 mg = ?

c. Volume that contains 75 mg = ?

d. Volume that contains 150 mg = ?

Slide 3 A car travels at 100 km/hour. Complete the table:

Distance (km)	Time (hour)
50	?
150	?
200	?

a. Time for 50 km = ?

b. Time for 150 km = ?

c. Time for 200 km = ?

Slide 4 The upper set is proportional to the lower set:

9	21	30	36
3	7	10	12

What is the ratio of the upper set to the lower set?

Slide 4 The upper set is proportional to the lower set:

2	5	7
10	25	35

What is the ratio of the upper set to the lower set?

Textbook Problems

Examples 2.1–2.5 · Page 7 In each of the following, decide whether or not the pair of sets is proportional.

2.1

6	10	14
3	5	7

2.2

5	7	9
40	56	72

2.3

2	4	8
1	2	3

2.4

5	7	9
15	17	19

2.5

16	17	21
80	85	105

Q1 · Page 26 Decide whether or not the pair of sets is proportional.

6	8	10…
3	5	7…

Proportional or not?

Q2 · Page 26 Decide whether or not the pair of sets is proportional.

5	7	9…
20	28	36…

Proportional or not?

Q3 · Page 26 Decide whether or not the pair of sets is proportional.

2	3	4…
5	6	7…

Proportional or not?

Q4 · Page 26 Decide whether or not the pair of sets is proportional.

2	4	6…
5	10	15…

Proportional or not?

Q5 · Page 26 Decide whether or not the pair of sets is proportional.

25	35…
5	7…

Proportional or not?

Q6 · Page 26 Decide whether or not the pair of sets is proportional.

2	4	6…
3	6	9…

Proportional or not?

1.2 Fractions

Lecture Problems

Slide 5 x/3 = 16/24

Find what is x.

Textbook Problems

Example 2.6 · Page 9 Find the numerator of the fraction that is equal to 1/7 but has denominator 21.

Numerator = ?

Example 2.7 · Page 9 Find the numerator of the fraction that is equal to 16/24 but has denominator 3. (Same as the lecture question on Slide 5.)

Numerator = ?

Example 2.8 · Page 9 Find the denominator of the fraction that is equal to 80/180, but has numerator 16.

Denominator = ?

Q7 · Page 26 Find the fraction that is equal to 7/8 but has denominator 72.

Answer = ?

Q8 · Page 26 Find the fraction that is equal to 4/7 but has numerator 28.

Answer = ?

Q9 · Page 26 Find the fraction that is equal to 2/9 but has denominator 117.

Answer = ?

Example 2.9 · Page 10 Express 35/45 in its lowest terms.

Lowest terms = ?

Example 2.10 · Page 10 Express 60/405 in its lowest terms.

Lowest terms = ?

Q10 · Page 26 Express 12/20 in its lowest terms.

Answer = ?

Q11 · Page 26 Express 3/15 in its lowest terms.

Answer = ?

Q12 · Page 26 Express 42/54 in its lowest terms.

Answer = ?

Q13 · Page 26 Express 78/130 in its lowest terms.

Answer = ?

Q14 · Page 26 Express 120/42 in its lowest terms.

Answer = ?

Q15 · Page 26 Multiply 3/13 by 5.

Answer = ?

Q16 · Page 26 Multiply 3/17 by 4.

Answer = ?

Q17 · Page 27 Multiply 5/9 by 7.

Answer = ?

1.3 Decimals

Lecture Problems

Slide 6 The number 0.23 is a rational number: the 2 is in the tenths column and the 3 is in the hundredths column, so 0.23 = 23/100.

a. Convert 0.5 to a fraction.

---

b. Convert 0.44 to a fraction.

Textbook Problems

Example 2.11 · Page 11 Convert 0.3 to a fraction.

Fraction = ?

Example 2.12 · Page 12 Convert 0.35 to a fraction.

Fraction = ?

Q18 · Page 27 Convert 0.8 to a fraction in its lowest terms.

Answer = ?

Q19 · Page 27 Convert 0.87 to a fraction in its lowest terms.

Answer = ?

Q20 · Page 27 Convert 0.875 to a fraction in its lowest terms.

Answer = ?

Example 2.13 · Page 12 Convert the fraction 3/5 to a decimal.

Decimal = ?

Example 2.14 · Page 12 Convert the fraction 7/8 to a decimal.

Decimal = ?

Q21 · Page 27 Convert the fraction 5/8 to a decimal.

Answer = ?

Q22 · Page 27 Convert the fraction 7/80 to a decimal.

Answer = ?

1.4 Percentages

Lecture Problems

Slide 8 8 percent = 8 per 100 = 8/100

a. Convert 3/8 to percent.

b. Convert 0.125 to percent.

c. Convert 25% to fraction.

d. Convert 0.6% to fraction.

Textbook Problems

Example 2.19 · Page 14 Express 7/10 as a percentage.

Answer = ?

Example 2.20 · Page 14 Express 3/8 as a percentage.

Answer = ?

Q27 · Page 27 Express 7/8 as a percentage.

Answer = ?

Q28 · Page 27 Express 6/25 as a percentage.

Answer = ?

Example 2.15 · Page 13 Express 0.76 as a percentage.

Answer = ?

Example 2.16 · Page 13 Express 0.769 as a percentage.

Answer = ?

Q23 · Page 27 Express 0.82 as a percentage.

Answer = ?

Q24 · Page 27 Express 0.345 as a percentage.

Answer = ?

Example 2.17 · Page 13 Express 20% as a fraction in its lowest terms.

Answer = ?

Example 2.18 · Page 13 Express 0.2% as a fraction in its lowest terms.

Answer = ?

Q25 · Page 27 Express 30% as a fraction in its lowest terms.

Answer = ?

Q26 · Page 27 Express 0.3% as a fraction in its lowest terms.

Answer = ?

1.5 Finding a Missing Value & Practical Problems

Textbook Problems

Example 2.21 · Page 16 For the proportional sets:

7	y	9	10…
21	24	27	30…

Find the missing value y.

Example 2.22 · Page 17 For the proportional sets:

x	14
25	50

Find the missing number x.

Example 2.23 · Page 18 For the proportional sets:

3	x
5	8

Find the missing value x.

Example 2.24 · Page 19 For the pair of proportional sets:

z	9
6	15

Find the missing value z.

Example 2.25 · Page 19 For the proportional sets:

37	52
7	z

Find the missing number z.

Example 2.28 · Page 21 Find the value of p in the two proportional sets:

36	9
22	p

p = ?

Q29 · Page 27 Find the missing value x from the pair of proportional sets:

8	x	20
2	4	5

Missing value = ?

Q30 · Page 27 Find the missing number from the pair of proportional sets:

25	30
y	6

Missing value = ?

Q31 · Page 27 Find the missing number from the pair of proportional sets:

21	z
7	30

Missing value = ?

Q32 · Page 28 Find the missing number from the pair of proportional sets:

81	18
x	2

Missing value = ?

Q33 · Page 28 Find the missing number from the pair of proportional sets:

y	21
8	24

Missing value = ?

Q34 · Page 28 Find the missing number from the pair of proportional sets:

4	10
10	z

Missing value = ?

Q35 · Page 28 Find the missing number from the pair of proportional sets:

x	10
7	35

Missing value = ?

Q36 · Page 28 Find the missing number from the pair of proportional sets:

32	y
19	41

Give your answer correct to two significant figures.

Missing value = ?

Q37 · Page 28 Find the missing number from the pair of proportional sets:

225	420
192	z

Give your answer correct to three significant figures.

Missing value = ?

Q38 · Page 28 Find the missing number from the pair of proportional sets:

x	11.3
22.5	15.7

Give your answer correct to three significant figures.

Missing value = ?

Example 2.26 · Page 20 Consider three sets each proportional to each other:

x	20
y	50
30	150

Find x and y.

Example 2.27 · Page 21 Find the values of a and b in the following proportional sets:

a	25	110
12	b	22

a = ?, b = ?

Example 2.29 · Page 21 Find the values of x and y in the three proportional sets:

24	6
x	15
64	y

x = ?, y = ?

Example 2.30 · Page 22 Find the values of y and z in the two proportional sets:

y	18	21
9	z	63

y = ?, z = ?

Worked example · Page 23 A suspension contains 120 mg of paracetamol in each 5 mL. Find the volume of suspension that contains 300 mg of paracetamol.

Volume = ?

Example 2.31 · Page 23 The recommended dose of phenytoin for a patient is 840 mg. An injection contains 50 mg/mL. What volume of injection is needed for the recommended dose?

Volume = ?

Example 2.32 · Page 24 1 L of aqueous solution contains 250 mL of syrup. How much syrup will be contained in 25 L of the solution?

Volume of syrup = ?

Example 2.33 · Page 24 25 g of ointment contains 5 g of calamine. How much calamine will be contained in 60 g of ointment?

Mass of calamine = ?

1.6 Exponentials (Powers of 10)

Lecture Problems

Slide 9 Physical and chemical measurements deal with either very small or large numbers, so for convenience we use exponentials. Positive power means multiply; negative power means divide.

a. Write 3,000,000 using a power of 10.

---

b. Write 0.000003 using a power of 10.

1.7 Ratio Strength, Percentage Strength & ppm

Lecture Problems

Slide 10 · Example 1 A percentage is the number of parts per one hundred and is probably the commonest expression of concentration used in pharmacy.

Express 1:2500 as a percentage strength.

Slide 11 · Example 2 Express 0.1% as a ratio strength.

Ratio strength = ?

Slide 11 · Example 3 Express 1 p.p.m. as a percentage strength.

Percentage strength = ?

1.8 Variation (Inverse Proportion)

Lecture Problems

Slide 12 Sometimes pharmaceutical calculations deal with inverse relationships — twice the cause, half the effect — as when you decrease the strength of a solution by increasing the amount of diluent.

If 10 litres of a 5% solution are diluted to 40 litres, what is the percentage strength of the dilution?

1.9 Units, Prefixes & Equivalences

Lecture Problems

Slide 14 Prefixes table — complete the **Factor** column:

Prefix	Symbol	Division	Factor
mega	M	× 1000,000	?
kilo	k	× 1000	?
deci	d	÷ 10	?
centi	c	÷ 100	?
milli	m	÷ 1000	?
micro	mc	÷ 1000,000	?
nano	n	÷ 1000,000,000	?

a. mega (M)

b. kilo (k)

c. deci (d)

d. centi (c)

e. milli (m)

f. micro (mc)

g. nano (n)

Slide 15 Complete the equivalences:

a. 1 L = ? ml

b. 1 kg = ? g = ? mg = ? mcg = ? ng

Slide 17 Examples — fill in the blanks:

a. 0.5 kg = ? g = ? mg

b. 0.25 g = ? mg (and in kg?)

c. 0.2 mg = ? mcg

d. 0.4 mg = ? kg

e. 0.5 L = ? ml

f. 0.25 mol = ? mmol

Slide 18 For drug calculation it is best to work in whole numbers (i.e. 125 mcg not 0.125 mg). To convert from a larger unit to a smaller unit: × 1000. From a smaller unit to a larger unit: ÷ 1000.

Write 0.125 mg as a whole number in a smaller unit.

Textbook Problems

Example 3.1 · Page 33 Change 200 mg to grams.

Answer = ?

Example 3.2 · Page 33 Convert 5473 mL to litres.

Answer = ?

Example 3.3 · Page 34 Convert 1.26 mg to micrograms.

Answer = ?

Example 3.4 · Page 34 Convert 135 465 micrograms to grams.

Answer = ?

Example 3.5 · Page 34 Convert 0.003 745 kL to mL.

Answer = ?

Q1 · Page 41 Change 0.035 g to milligrams.

Answer = ?

Q2 · Page 41 Change 1384 mg to grams.

Answer = ?

Q3 · Page 41 Change 437 mL to litres.

Answer = ?

Q4 · Page 42 Change 12.47 L to millilitres.

Answer = ?

Q5 · Page 42 Change 0.87 mL to microlitres.

Answer = ?

Q6 · Page 42 Change 15 750 microlitres to millilitres.

Answer = ?

Q7 · Page 42 Change 0.0025 kg to grams.

Answer = ?

Q8 · Page 42 Change 12 g to kilograms.

Answer = ?

Q9 · Page 42 Change 0.72 megalitres to kilolitres.

Answer = ?

Q10 · Page 42 Change 145 kg to megagrams.

Answer = ?

1.10 Conversions Between Systems & Temperature

Lecture Problems

Slide 19 (practice) Height: convert inch to cm. Weight: convert pound to kg. (The slide gives no numbers — these two use the standard factors 1 inch = 2.54 cm and 1 pound = 0.4536 kg.)

a. Convert 10 inches to cm.

---

b. Convert 150 pounds to kg.

Textbook Problems

Example 3.6 · Page 36 Convert 3.4 pints to litres. (1 L = 1.76 pints)

Litres = ?

Q11 · Page 42 Use the conversion factor 1 L = 1.76 pints to express:

(a) 6.43 pints in litres

(b) 2.42 L in pints

Example 3.7 · Page 37 Convert 106.4 kg to stones and pounds. (1 pound = 0.4536 kg; 14 pounds = 1 stone)

Stones and pounds = ?

Example 3.8 · Page 37 Convert 10 stone 4 pounds to kilograms. (1 pound = 0.4536 kg)

Kilograms = ?

Q12 · Page 42 Use the conversion factor 1 pound = 0.4536 kg to express:

(a) 9 stones 4 pounds in kilograms (1 stone = 14 pounds)

(b) 49 kg in stones and pounds

Example 3.9 · Page 38 Convert 5 feet 10 inches into metres. (1 inch = 25.4 mm = 0.0254 m)

Metres = ?

Example 3.10 · Page 38 What would be the BMI for a patient who weighs 10 stone and 4 pounds and is 5 feet 10 inches tall? (BMI = weight (kg) / height² (m))

BMI = ?

Worked example · Page 40 Find the number of degrees Celsius corresponding to 95 degrees Fahrenheit. (50 °C = 122 °F; water freezes at 0 °C = 32 °F)

°C = ?

Example 3.11 · Page 40 Change 25 degrees Celsius to degrees Fahrenheit. (0–100 °C corresponds to 32–212 °F, a range of 100 °C and 180 °F)

°F = ?

Example 3.12 · Page 41 Change 68 degrees Fahrenheit to degrees Celsius.

°C = ?

Q13 · Page 42 The boiling point of water is 100 degrees Celsius or 212 degrees Fahrenheit. The freezing point of water is 0 degrees Celsius or 32 degrees Fahrenheit. Express:

(a) 15 degrees Celsius in degrees Fahrenheit (to the nearest degree)

(b) 88 degrees Fahrenheit in degrees Celsius (to the nearest degree)

1.11 Dosage Calculations

Lecture Problems

Slide 21 The dose required is 3 mg/kg and the patient weighs 68 kg. This means that for every kg of patient's weight, you will need 3 mg of drug.

Total dose required = ?

Slide 22 In some cases, the dose is given as a total daily dose — it has to be given in divided doses (3 or 4 times a day).

Ibuprofen 1.2 g daily given 3 times daily. What is each single dose?

Slide 23 The dose for Cefuroxime iv is 2.25 g in divided doses every 8 hours.

What is each dose?

Slide 24 You need to prepare an infusion of digoxin containing 0.75 mg. Digoxin is available as 500 mcg/2 ml.

How many ml do you need?

Answers · Lecture 1

1.1 Proportional Sets & Ratios

Slide 3

a. 1 ml — 100 mg → 5 ml
x = 20 × 5 / 100 = 1 ml

b. 2.5 ml — x = 50 × 5 / 100 = 2.5 ml

c. 3.75 ml — x = 75 × 5 / 100 = 3.75 ml

d. 7.5 ml — x = 150 × 5 / 100 = 7.5 ml

Slide 3

a. 1/2 hour — Time = 50 / 100 = 1/2 hour

b. 1.5 hours — Time = 150 / 100 = 1.5 hours

c. 2 hours — Time = 200 / 100 = 2 hours

Slide 4

3 : 1 — Each number in the upper set is (×3) the corresponding number in the lower set:
9 = 3 × 3, 21 = 7 × 3, 30 = 10 × 3, 36 = 12 × 3
Ratio = 3 : 1

Slide 4

1 : 5 — 10 = 2 × 5, 25 = 5 × 5, 35 = 7 × 5
So upper : lower = 1 to 5 = 1 : 5

Examples 2.1–2.5 · Page 7

2.1 Proportional — Each value of the upper set is twice the corresponding value of the lower set (6 = 2 × 3, 10 = 2 × 5, 14 = 2 × 7).

2.2 Proportional — Each value of the lower set is eight times the corresponding value of the upper set (40 = 8 × 5, 56 = 8 × 7, 72 = 8 × 9).

2.3 Not proportional — 2 ÷ 1 = 2, 4 ÷ 2 = 2, but 8 ÷ 3 ≈ 2.67 — there is no single fixed number.

2.4 Not proportional — 15 ÷ 5 = 3, but 17 ÷ 7 ≈ 2.43 — no fixed multiplier. (Here 10 is added each time, which is not proportion.)

2.5 Proportional — Each value of the lower set is five times the corresponding value of the upper set (80 = 5 × 16, 85 = 5 × 17, 105 = 5 × 21).

Q1 · Page 26

Not proportional — 6 ÷ 3 = 2, but 8 ÷ 5 = 1.6 — no fixed multiplier.

Q2 · Page 26

Proportional — Each value of the lower set is four times the corresponding value of the upper set (20 = 4 × 5, 28 = 4 × 7, 36 = 4 × 9).

Q3 · Page 26

Not proportional — 5 ÷ 2 = 2.5, but 6 ÷ 3 = 2 — no fixed multiplier.

Q4 · Page 26

Proportional — Each number of the lower set is 2.5 times the corresponding value of the upper set (5 = 2.5 × 2, 10 = 2.5 × 4, 15 = 2.5 × 6).

Q5 · Page 26

Proportional — Each value of the upper set is 5 times the corresponding value of the lower set (25 = 5 × 5, 35 = 5 × 7).

Q6 · Page 26 **Proportional** — Each value of the lower set is 1.5 times the corresponding value of the upper set (3 = 1.5 × 2, 6 = 1.5 × 4, 9 = 1.5 × 6).

1.2 Fractions

Slide 5 **x = 2** — Denominator 24 has been divided by 8 to get 3, so numerator 16 must also be divided by 8:
x = 16 ÷ 8 = 2
Check: 2/3 = 16/24 ✓

Example 2.6 · Page 9 **3** — Let the numerator be x: 1/7 = x/21
Denominator 7 has been multiplied by 3 to get 21, so numerator 1 must be multiplied by 3:
x = 3

Example 2.7 · Page 9 **2** — Let the numerator be y: y/3 = 16/24
Denominator 24 has been divided by 8 to get 3, so numerator 16 must be divided by 8:
y = 2

Example 2.8 · Page 9 **36** — Let the denominator be x: 16/x = 80/180
Numerator 80 has been divided by 5 to get 16, so x is 180 divided by 5:
x = 36

Q7 · Page 26 **63/72** — 8 × 9 = 72, so the numerator is 7 × 9 = 63.
7/8 = 63/72

Q8 · Page 26 **28/49** — 4 × 7 = 28, so the denominator is 7 × 7 = 49.
4/7 = 28/49

Q9 · Page 26 **26/117** — 9 × 13 = 117, so the numerator is 2 × 13 = 26.
2/9 = 26/117

Example 2.9 · Page 10 **7/9** — Both 35 and 45 can be divided by 5:
35/45 = 7/9

Example 2.10 · Page 10 **4/27** — 60 and 405 will divide by 5: 60/405 = 12/81
12 and 81 will divide by 3: 12/81 = 4/27

Q10 · Page 26 **3/5** — Divide top and bottom by 4: 12/20 = 3/5

Q11 · Page 26 **1/5** — Divide top and bottom by 3: 3/15 = 1/5

Q12 · Page 26 **7/9** — Divide top and bottom by 6: 42/54 = 7/9

Q13 · Page 26 **3/5** — Divide by 2: 78/130 = 39/65
Divide by 13: 39/65 = 3/5

Q14 · Page 26 **20/7 = 2 6/7** — Divide top and bottom by 6: 120/42 = 20/7
20/7 = 2 6/7 (2 whole and 6/7)

Q15 · Page 26 **15/13** — Multiply only the numerator: 3 × 5 = 15
3/13 × 5 = 15/13

Q16 · Page 26 **12/17** — 3 × 4 = 12, so 3/17 × 4 = 12/17

Q17 · Page 27 **35/9 = 3 8/9** — 5 × 7 = 35, so 5/9 × 7 = 35/9
35/9 = 27/9 + 8/9 = 3 8/9

1.3 Decimals

Slide 6
a. **1/2** — The 5 is in the tenths column, so 0.5 = 5/10
Lowest terms: 5/10 = 1/2
b. **11/25 (= 44/100)** — The last 4 is in the hundredths column, so 0.44 = 44/100
Lowest terms (÷4): 44/100 = 11/25

Example 2.11 · Page 11 **3/10** — The 3 is in the tenths column, so 0.3 = 3/10

Example 2.12 · Page 12 **7/20** — The 5 is in the hundredths column, so 0.35 = 35/100
Lowest terms (÷5): 35/100 = 7/20

Q18 · Page 27 **4/5** — 0.8 = 8/10
Lowest terms (÷2): 8/10 = 4/5

Q19 · Page 27 **87/100** — 0.87 = 87/100
87 = 3 × 29 and 100 = 2 × 2 × 5 × 5 — nothing in common, so it is already in lowest terms.

Q20 · Page 27 **7/8** — 0.875 = 875/1000
÷125: 875/1000 = 7/8

Example 2.13 · Page 12 **0.6** — 3 divided by 5 gives 0.6, or:
3/5 = 6/10 = 0.6

Example 2.14 · Page 12 **0.875** — 7 divided by 8 gives 0.875

Q21 · Page 27 **0.625** — 5 ÷ 8 = 0.625

Q22 · Page 27 **0.0875** — 7 ÷ 80 = 0.0875

1.4 Percentages

Slide 8

a. 37.5% — 3/8 = 3 ÷ 8 = 0.375
0.375 × 100 = 37.5%

b. 12.5% — 0.125 × 100 = 12.5%

c. 1/4 — 25% = 25/100
Lowest terms (÷25): 25/100 = 1/4

d. 3/500 — 0.6% = 0.6/100 = 6/1000
Lowest terms (÷2): 6/1000 = 3/500

Example 2.19 · Page 14 **70%** — 7/10 = 0.70 = 70%

Example 2.20 · Page 14 **37.5%** — 3/8 = 0.375 = 37.5% (Same as the lecture question on Slide 8.)

Q27 · Page 27 **87.5%** — 7/8 = 7 ÷ 8 = 0.875
0.875 × 100 = 87.5%

Q28 · Page 27 **24%** — 6/25 = 24/100 = 24%
(or 6 ÷ 25 = 0.24 = 24%)

Example 2.15 · Page 13 **76%** — There is a 6 in the hundredths column:
0.76 = 76 hundredths = 76%

Example 2.16 · Page 13 **76.9%** — 0.769 = 76.9 hundredths = 76.9%

Q23 · Page 27 **82%** — 0.82 × 100 = 82%

Q24 · Page 27 **34.5%** — 0.345 × 100 = 34.5%

Example 2.17 · Page 13 **1/5** — 20% = 20/100 = 1/5

Example 2.18 · Page 13 **1/500** — 0.2% = 0.2/100 = 2/1000 = 1/500

Q25 · Page 27 **3/10** — 30% = 30/100 = 3/10

Q26 · Page 27 **3/1000** — 0.3% = 0.3/100 = 3/1000

1.5 Finding a Missing Value & Practical Problems

Example 2.21 · Page 16 **y = 8** — Method A: y/24 = 9/27
24 × y/24 = 24 × 9/27 → y = 216/27 = 8
Method B (spotting): each lower value is 3 times the upper value, so y × 3 = 24 → y = 8

Example 2.22 · Page 17 **x = 7** — Method C: the vertical sets are proportional. The ratio of 50 to 25 is 2, so the ratio of 14 to x is also 2:
x = 14 ÷ 2 = 7
Method A: x/25 = 14/50 → x = 25 × 14/50 = 350/50 = 7

Example 2.23 · Page 18 **x = 24/5 = 4.8** — Method A: x/8 = 3/5
x = 8 × 3/5 = 24/5
x = 4 4/5 = 4.8

Example 2.24 · Page 19 **z = 3.6** — z/6 = 9/15
z = 6 × 9/15 = 54/15 = 3.6

Example 2.25 · Page 19 **z = 9.84** — Write the equation with z on top: z/52 = 7/37
z = 52 × 7/37 = 364/37 = 9.837837…
Correct to three significant figures: z = 9.84

Example 2.28 · Page 21 **p = 5.5** — Spotting (Method C): the ratio of 36 to 9 is 4, so 22 is divided by 4:
p = 22 ÷ 4 = 5.5

Q29 · Page 27 **x = 16** — Each upper value is 4 times the lower value (8 = 4 × 2, 20 = 4 × 5)
x = 4 × 4 = 16

Q30 · Page 27 **y = 5** — Upper = 5 × lower (30 = 5 × 6)
y = 25 ÷ 5 = 5

Q31 · Page 27 **z = 90** — Upper = 3 × lower (21 = 3 × 7)
z = 3 × 30 = 90

Q32 · Page 28 **x = 9** — The ratio of 18 to 2 is 9, so 81 ÷ 9:
x = 9

Q33 · Page 28 **y = 7** — Across the rows: 24 = 3 × 8, so 21 = 3 × y
y = 21 ÷ 3 = 7

Q34 · Page 28 **z = 25** — Lower = 2.5 × upper (10 = 2.5 × 4)
z = 2.5 × 10 = 25
(Method A: z/10 = 10/4 → z = 100/4 = 25)

Q35 · Page 28 **x = 2** — Lower = 3.5 × upper (35 = 3.5 × 10), or across: 35 = 5 × 7, so 10 = 5 × x
x = 10 ÷ 5 = 2

Q36 · Page 28 **y = 69** — Method A: y/41 = 32/19
y = 41 × 32/19 = 1312/19 = 69.05…
Correct to two significant figures: y = 69

Q37 · Page 28 **z = 358** — Method A: z/420 = 192/225
z = 420 × 192/225 = 80640/225 = 358.4
Correct to three significant figures: z = 358

Q38 · Page 28 **x = 16.2** — Method A: x/22.5 = 11.3/15.7
x = 22.5 × 11.3/15.7 = 254.25/15.7 = 16.19…
Correct to three significant figures: x = 16.2

Example 2.26 · Page 20 **x = 4, y = 10** — Method A: x/30 = 20/150 → x = 600/150 = 4
y/30 = 50/150 → y = 1500/150 = 10
Spotting (Method C): the ratio of 150 to 30 is 5, so x = 20 ÷ 5 = 4 and y = 50 ÷ 5 = 10

Example 2.27 · Page 21 **a = 60, b = 5** — Spotting (Method B): the ratio of 110 to 22 is 5
a = 5 × 12 = 60
b = 25 ÷ 5 = 5

Example 2.29 · Page 21 **x = 60, y = 16** — Spotting (Method C): the ratio of 24 to 6 is 4
x = 15 × 4 = 60
y = 64 ÷ 4 = 16

Example 2.30 · Page 22 **y = 3, z = 54** — Spotting (Method B): the ratio of 63 to 21 is 3
y = 9 ÷ 3 = 3
z = 18 × 3 = 54

Worked example · Page 23 **12.5 mL** — Let the volume be x mL:

volume of suspension (mL)	x	5
amount of paracetamol (mg)	300	120

x/300 = 5/120
x = 300 × 5/120 = 12.5 mL

Example 2.31 · Page 23 **16.8 mL** — Let the volume be y mL:

amount of phenytoin (mg)	840	50
volume of injection (mL)	y	1

y/840 = 1/50
y = 840/50 = 16.8 mL

Example 2.32 · Page 24 **6250 mL (= 6.25 L)** — Let the syrup be x mL:

volume of syrup (mL)	250	x
volume of solution (L)	1	25

x = 250 × 25 = 6250 mL = 6.25 L

Example 2.33 · Page 24 **12 g** — Let the calamine be y g:

mass of calamine (g)	5	y
mass of ointment (g)	25	60

y/60 = 5/25
y = 60 × 5/25 = 12 g

1.6 Exponentials (Powers of 10)

Slide 9

a. **3×10^{6}** — 3,000,000 = 3 × 1,000,000 = 3 × 10^{6} (positive power)

b. **3×10^{-6}** — 0.000003 = 3 / 1,000,000 = 3 × 10^{-6} (minus power)

1.7 Ratio Strength, Percentage Strength & ppm

Slide 10 · Example 1

0.04% — 1 part / 2500 parts = x parts / 100 parts

x = 1 × 100 / 2500 = 0.04

Percentage strength = 0.04%

Slide 11 · Example 2

1:1000 — 0.1 g / 100 g = 1 part / y parts

y = 100 × 1 / 0.1 = 1000

Ratio strength = 1:1000

Slide 11 · Example 3

0.0001% — 1 p.p.m. is 1 part per million = 1:1,000,000

1 part / 1,000,000 = y parts / 100 parts

y = 1 × 100 / 1,000,000 = 0.0001%

1.8 Variation (Inverse Proportion)

Slide 12

1.25% — Method 1: 10 (litres) / 40 (litres) = x(%) / 5(%)

x = 10 × 5 / 40 = 1.25%

Method 2: $C_1V_1 = C_2V_2$

5 × 10 = C_2 × 40

C_2 = 50 / 40 = 1.25%

1.9 Units, Prefixes & Equivalences

Slide 14

a. **10^{6}** — × 1,000,000 = 10^{6}

b. **10^{3}** — × 1000 = 10^{3}

c. **10^{-1}** — ÷ 10 = 10^{-1}

d. **10^{-2}** — ÷ 100 = 10^{-2}

e. **10^{-3}** — ÷ 1000 = 10^{-3}

f. **10^{-6}** — ÷ 1,000,000 = 10^{-6}

g. **10^{-9}** — ÷ 1,000,000,000 = 10^{-9}

Slide 15

a. **1000 ml** — 1 L = 1000 ml

b. **1000 g = 1,000,000 mg = 1,000,000,000 mcg = 1,000,000,000,000 ng** — 1 kg = 1000 g

1000 g × 1000 = 1,000,000 mg

1,000,000 mg × 1000 = 1,000,000,000 mcg

1,000,000,000 mcg × 1000 = 1,000,000,000,000 ng

Slide 17

a. 500 g = 500,000 mg — 0.5 × 1000 = 500 g
500 × 1000 = 500,000 mg

b. 250 mg = 0.00025 kg — 0.25 g × 1000 = 250 mg
0.25 g ÷ 1000 = 0.00025 kg

c. 200 mcg — 0.2 × 1000 = 200 mcg

d. 0.0000004 kg — 0.4 mg ÷ 1000 = 0.0004 g
0.0004 g ÷ 1000 = 0.0000004 kg

e. 500 ml — 0.5 × 1000 = 500 ml

f. 250 mmol — 0.25 × 1000 = 250 mmol

Slide 18

125 mcg — 0.125 mg × 1000 = 125 mcg

Example 3.1 · Page 33

0.2 g — g – – mg
200 mg = 0 2 0 0 = 0.2 g
(or 200 ÷ 1000 = 0.2 g)

Example 3.2 · Page 33

5.473 L — L – – mL
5473 mL = 5 4 7 3 = 5.473 L
(or 5473 ÷ 1000 = 5.473 L)

Example 3.3 · Page 34

1260 micrograms — mg – – micrograms
1.26 mg = 1 2 6 0 = 1260 micrograms
(or 1.26 × 1000 = 1260)

Example 3.4 · Page 34

0.135 465 g — g – – mg – – micrograms
135 465 micrograms = 0 1 3 5 4 6 5 = 0.135 465 g
(or 135 465 ÷ 1,000,000 = 0.135 465 g)

Example 3.5 · Page 34

3745 mL — kL – – L – – mL
0.003 745 kL = 0 0 0 3 7 4 5 = 3745 mL
(or 0.003745 × 1,000,000 = 3745 mL)

Q1 · Page 41

35 mg — 0.035 × 1000 = 35 mg

Q2 · Page 41

1.384 g — 1384 ÷ 1000 = 1.384 g

Q3 · Page 41

0.437 L — 437 ÷ 1000 = 0.437 L

Q4 · Page 42

12 470 mL — 12.47 × 1000 = 12 470 mL

Q5 · Page 42

870 microlitres — 0.87 × 1000 = 870 microlitres

Q6 · Page 42

15.75 mL — 15 750 ÷ 1000 = 15.75 mL

Q7 · Page 42

2.5 g — 0.0025 × 1000 = 2.5 g

Q8 · Page 42

0.012 kg — 12 ÷ 1000 = 0.012 kg

Q9 · Page 42 **720 kL** — 1 megalitre = 1000 kL
0.72 × 1000 = 720 kL

Q10 · Page 42 **0.145 megagrams** — 1 megagram = 1000 kg
145 ÷ 1000 = 0.145 megagrams

1.10 Conversions Between Systems & Temperature

Slide 19 (practice) **a. 25.4 cm** — 1 inch = 2.54 cm
10 × 2.54 = 25.4 cm

b. 68.04 kg — 1 pound = 0.4536 kg
150 × 0.4536 = 68.04 kg

Example 3.6 · Page 36 **1.93 L** — Let the required number of litres be x:

litres	x	1
pints	3.4	1.76

x/3.4 = 1/1.76
x = 3.4 / 1.76 = 1.93
3.4 pints = 1.93 L

Q11 · Page 42 **(a) 3.65 L** — x = 6.43 / 1.76 = 3.65 L
(b) 4.26 pints — 2.42 × 1.76 = 4.26 pints

Example 3.7 · Page 37 **16 stone 11 pounds** — Let the number of pounds be y:

pounds	y	1
kilograms	106.4	0.4536

y/106.4 = 1/0.4536
y = 106.4 / 0.4536 = 234.56 ≈ 235 pounds
235 ÷ 14 = 16 stone with 11 pounds left (16 × 14 = 224; 235 − 224 = 11)
106.4 kg = 16 stone 11 pounds

Example 3.8 · Page 37 **65.3 kg** — 14 pounds = 1 stone, so 10 stone 4 pounds = 140 + 4 = 144 pounds

pounds	1	144
kilograms	0.4536	x

x/144 = 0.4536/1
x = 144 × 0.4536 = 65.3 kg

Q12 · Page 42 **(a) 59 kg** — 9 stones 4 pounds = 9 × 14 + 4 = 130 pounds
130 × 0.4536 = 58.97 ≈ 59 kg

(b) 7 stone 10 pounds — 49 / 0.4536 = 108.02 ≈ 108 pounds
108 ÷ 14 = 7 stone (7 × 14 = 98) with 10 pounds left
49 kg = 7 stone 10 pounds

Example 3.9 · Page 38 **1.778 m** — 12 inches = 1 foot, so 5 feet 10 inches = 60 + 10 = 70 inches

inches	1	70
metres	0.0254	x

x/70 = 0.0254/1
x = 70 × 0.0254 = 1.778 m

Example 3.10 · Page 38 **20.7** — From Example 3.8: 10 stone 4 pounds = 65.3 kg
From Example 3.9: 5 feet 10 inches = 1.778 m
BMI = 65.3 / (1.778)² = 65.3 / (1.778 × 1.778) = 65.3 / 3.161 = 20.7

Worked example · Page 40 **35 °C** — Celsius and Fahrenheit are NOT proportional — subtract 32 from the Fahrenheit values first:

degrees Celsius	50	y
degrees Fahrenheit minus 32	122 − 32 = 90	95 − 32 = 63

y/63 = 50/90
y = 63 × 50 / 90 = 35
95 °F = 35 °C

Example 3.11 · Page 40 **77 °F** —

degrees Fahrenheit minus 32	180	x − 32
degrees Celsius	100	25

Spotting: 100 divided by 4 is 25, so 180 divided by 4 is x − 32:
x − 32 = 45
x = 45 + 32 = 77
25 °C = 77 °F

Example 3.12 · Page 41 **20 °C** —

degrees Fahrenheit minus 32	180	68 − 32 = 36
degrees Celsius	100	y

y/36 = 100/180
y = 100 × 36 / 180 = 20
68 °F = 20 °C

Q13 · Page 42 **(a) 59 °F** — (x − 32)/180 = 15/100
x − 32 = 180 × 15 / 100 = 27
x = 27 + 32 = 59 °F
(b) 31 °C — 88 − 32 = 56
y/56 = 100/180
y = 100 × 56 / 180 = 31.1 ≈ 31 °C

1.11 Dosage Calculations

Slide 21 **204 mg** — Total dose required = dose/kg × patient's weight
= 3 × 68 = 204 mg

Slide 22 **400 mg** — 1.2 g = 1200 mg
1200 / 3 = 400 mg for each single dose

Slide 23 **750 mg every 8 hours** — 8 hourly = 24 ÷ 8 = 3 divided doses daily
2.25 / 3 = 0.75 g (i.e. 750 mg)
So, the dose for cefuroxime iv is 750 mg every 8 hours

Slide 24 **3 ml** — Convert everything to mcg: 0.75 mg = 0.75 × 1000 = 750 mcg
500 mcg → 2 ml
750 mcg → x
x = 750 × 2 / 500 = 3 ml
(Amount you want / amount you have × volume)

2 Drug Concentrations

In this lecture

1. Fractions, Decimals & Percentages 2 lecture
2. Calculations with % 3 lecture
3. Ratio Strength 1 lecture · 7 textbook
4. Percentage Concentration (w/v, w/w, v/v) 3 lecture · 16 textbook
5. mg/ml Concentrations & Amount Strength 2 lecture · 2 textbook
6. Parts per Million (ppm) 2 textbook
7. Converting Between Expressions of Concentration 6 textbook
8. Heparin & Insulin (Units) 1 lecture
9. Practice Calculations (Lecture) 3 lecture

2.1 Fractions, Decimals & Percentages

Lecture Problems

Slide 5 Fractions & percentages: to convert a fraction to a percentage, multiply by 100; to convert a percentage to a fraction, divide by 100.

a. Convert 2/8 to a percentage.

b. Convert 25% to a fraction.

Slide 6 Decimals & percentages: to convert a decimal to a percentage, multiply by 100; to convert a percentage to a decimal, divide by 100.

a. Convert 0.5 to a percentage.

b. Convert 50% to a decimal.

2.2 Calculations with %

Lecture Problems

Slide 7 How much is 35% in 500 ml?

Amount = ?

Slide 8 What is the percentage of 480 of 8000?

Percentage = ?

Slide 9 How many ml is 40% of 1.3 litre?

Volume (ml) = ?

2.3 Ratio Strength

Lecture Problems

Slide 10 Ratio strength is expressed as a ratio in the form of 1 in x.

5 g of product contains 250 mg of sulphur in yellow soft paraffin. Express this in ratio.

Textbook Problems

Example 4.2 · Page 49 2 L of an aqueous solution contains 50 mL of ethanol. Express this as a ratio strength.

Ratio strength = ?

Example 4.3 · Page 49 5 g of product contains 250 mg of sulfur in yellow soft paraffin. Express this as a ratio strength. (Same as the lecture question on Slide 10.)

Ratio strength = ?

Example 4.16 · Page 60 What is the amount of potassium permanganate in 300 mL of a 1 in 25 solution and what is the percentage strength of the solution?

Amount = ? and % = ?

Q4 · Page 61 How many milligrams of y are needed to make 200 mL of a 1 in 500 solution?

Answer = ?

Q5 · Page 61 How many millilitres of y are needed to produce 400 mL of a 1 in 200 solution?

Answer = ?

Q6 · Page 61 How many milligrams of y are needed to produce 25 g of a 1 in 5 ointment?

Answer = ?

Q7 · Page 61 How many grams of y are there in 250 mL of a 1 in 80 solution?

Answer = ?

2.4 Percentage Concentration (w/v, w/w, v/v)

Lecture Problems

Slide 11 The most common way of describing drug concentration is percentage concentration.

a. What does 3% w/v mean?

---

b. What does 3% w/w mean?

---

c. What does 3% v/v mean?

Slide 12 How much dextrose is there in a 500 ml infusion of dextrose 4% w/v?

Dextrose = ?

Slide 13 You have salbutamol nebuliser solution 0.3%.

What volume is required for a 4 mg dose?

Textbook Problems

Example 4.15 · Page 59 How many milligrams of aluminium acetate are required to prepare 500 mL of a 0.03% w/v solution?

Amount (mg) = ?

Q9 · Page 62 How many milligrams of x are needed to make 5 mL of an 8% solution?

Answer = ?

Q10 · Page 62 How many millilitres of x are needed to make 600 mL of a 3% solution?

Answer = ?

Q11 · Page 62 How many milligrams of x are needed to make 200 mL of a 3.4% solution?

Answer = ?

Q12 · Page 62 How many milligrams of x are there in 100 mL of a 0.01% solution?

Answer = ?

Q13 · Page 62 How much calamine is required to produce 250 g of a 3% ointment?

Answer = ?

Q8 · Page 62 Potassium permanganate 0.1 g; Purified water to 100 mL.

How many millilitres of a 2.5% potassium permanganate solution could be used in place of the 0.1 g in the above formula?

Q14 · Page 62 What volume of 17% w/v solution contains 1.5 g of ingredient?

Answer = ?

Q15 · Page 62 What volume of 20% w/v solution contains 5 g of ingredient?

Answer = ?

Q16 · Page 62 What volume of 15% w/v solution contains 7 g of ingredient?

Answer = ?

Q17 · Page 62 What volume of 34% w/v solution contains 150 g of ingredient?

Answer = ?

Example 4.6 · Page 52 A cream contains 12 g of drug X made up to 100 g with cream base. What is the percentage concentration?

Percentage = ?

Example 4.8 · Page 53 Express 900 mg of sodium chloride made up to 100 mL with water as a percentage.

Percentage = ?

Q3 · Page 61 If the following amounts of drug are made up to 5 g with lactose what is the percentage concentration of the resulting mix?

(a) 100 mg

(b) 150 mg

(c) 200 mg

(d) 300 mg

(e) 500 mg

Example 4.7 · Page 52 Express 1 in 500 w/v solution of potassium permanganate as a percentage.

Percentage = ?

Example 4.9 · Page 53 A morphine sulfate injection contains 10 mg/mL. What is the percentage concentration?

Percentage = ?

2.5 mg/ml Concentrations & Amount Strength

Lecture Problems

Slide 14 Sometimes it may be useful to convert percentage concentrations to mg/ml.

Express sodium chloride 0.9% in mg/ml.

Slide 16 You can find the strength in mg/ml by multiplying the percentage by 10; to convert mg/ml to percentage, divide by 10.

a. Convert 0.9% to mg/ml.

b. Convert 9 mg/ml to a percentage.

Textbook Problems

Example 4.1 · Page 47 A preparation contains 900 mg of sodium chloride dissolved in water to produce 100 mL of solution. Express the concentration of the solution as an amount strength (mg/mL and g/mL).

Amount strength = ?

Example 4.12 · Page 56 A solution contains a solid dissolved in a liquid. The ratio strength is 1 in 2000 w/v. What are the percentage strength and the amount concentration expressed as mg/mL?

% = ? and mg/mL = ?

2.6 Parts per Million (ppm)

Textbook Problems

Example 4.4 · Page 50 Fluoride supplements should not be taken if the amount of fluoride in the water supply exceeds 0.7 parts per million w/v. Express this ratio in mg/L.

mg/L = ?

Example 4.5 · Page 51 If the concentration of fluoride is 0.25 ppm w/v, how many litres would contain 1 mg of fluoride?

Litres = ?

2.7 Converting Between Expressions of Concentration

Textbook Problems

Example 4.10 · Page 54 A solution contains 20 mL of ethanol in 500 mL of product. Express the concentration as a ratio strength and as a percentage strength.

Ratio = ? and % = ?

Example 4.11 · Page 55 A solid ingredient mixed with a solid vehicle has a ratio strength of 1 in 40. Find the percentage strength and the amount strength expressed as grams per gram.

% = ? and g/g = ?

Q1 · Page 60 Convert the following ratio strengths into percentage strengths:

(a) 1 in 25

(b) 1 in 20

(c) 1 in 50

(d) 1 in 800

(e) 1 in 500

(f) 1 in 2000

(g) 1 in 300

Example 4.13 · Page 57 A liquid ingredient mixed with another liquid vehicle has a concentration of 5% v/v. Find the ratio strength and the amount strength expressed as mL/mL.

Ratio = ? and mL/mL = ?

Example 4.14 · Page 57 5 g of solid ingredient is added to 45 g of a base. Find the percentage strength, the ratio strength and the amount strength expressed as g/g. (Same as the lecture question on Slide 22.)

% = ?, ratio = ?, g/g = ?

Q2 · Page 61 What is the concentration of the solutions, expressed as percentage strength and ratio strength, when the following amounts of drug Z are dissolved in enough water to produce 125 mL of solution?

(a) 25 g

(b) 50 g

(c) 60 g

(d) 5 g

(e) 7 g

2.8 Heparin & Insulin (Units)

Lecture Problems

Slide 18 Heparin is available as 1000, 5000, 10000 and 25000 units/ml (1 ml ampules).

You need to give heparin as continuous infusion 13000 units in 35 ml normal saline over 24 hours. What do you withdraw?

2.9 Practice Calculations (Lecture)

Lecture Problems

Slide 20 Practice Calculation 1

How many grams of sodium chloride are there in one litre infusion of sodium chloride 0.9%?

Slide 21 Practice Calculation 2

You need to give infusion of calcium gluconate 3 g. You have 10 ml ampules of calcium gluconate 10%. How much do you need to withdraw?

Slide 22 Practice Calculation 3

5 g of solid ingredient is added to 45 g of a base. Express as percentage, ratio, and as amount of strength g/g.

Answers · Lecture 2

2.1 Fractions, Decimals & Percentages

Slide 5

a. 25% — 2/8 = 1/4
1/4 × 100 = 25%

b. 1/4 — 25% = 25/100 = 5/20 = 1/4 (= 2/8)

Slide 6

a. 50% — 0.5 × 100 = 50%

b. 0.5 — 50% = 50/100 = 0.5

2.2 Calculations with %

Slide 7

175 ml — Assume 500 = 100%
500 → 100
x → 35
x = 500 × 35 / 100 = 175 ml
OR use the formula: Number / 100 × % required = 500 / 100 × 35 = 175 ml

Slide 8

6% — Assume 8000 is 100%
8000 → 100
480 → x
x = 480 × 100 / 8000 = 6%
OR use the formula: smaller number / larger number × 100 = 480 / 8000 × 100 = 6%

Slide 9

520 ml — a) Convert litre to ml (because the required answer is in ml): 1.3 litre = 1.3 × 1000 = 1300 ml
b) Consider 1300 ml as 100%
c) Cross multiply: 1300 → 100, x → 40
x = 1300 × 40 / 100 = 520 ml

2.3 Ratio Strength

Slide 10

1 in 20 (w/w) — This is w/w — must be converted to the same units first:
250 mg = 0.25 g
0.25 → 5
1 → x
x = 5 × 1 / 0.25 = 20
Ratio strength = 1 in 20

Example 4.2 · Page 49 **1 in 40 v/v** — v/v — convert to the same units: 2 L = 2000 mL

ethanol (mL)	50	1
product (mL)	2000	r

By spotting: 50 ÷ 50 = 1, so r = 2000 ÷ 50 = 40
Ratio strength = 1 in 40 v/v

Example 4.3 · Page 49 **1 in 20 w/w** — Both solid → w/w. Convert: 250 mg = 0.25 g

sulfur (g)	0.25	1
product (g)	5	r

5/0.25 = r/1 → r = 5/0.25 = 20
Ratio strength = 1 in 20 w/w

Example 4.16 · Page 60 **12 g; 4% w/v** — 1 in 25 means 1 g in 25 mL

	ratio	amount in 300 mL	percentage
potassium permanganate (g)	1	a	p
product (mL)	25	300	100

a/300 = 1/25 → a = 300/25 = 12 g
By spotting: 25 × 4 = 100, so p = 1 × 4 = 4
12 g in 300 mL; 4% w/v

Q4 · Page 61 **400 mg** — 1 in 500 = 1 g in 500 mL
a/200 = 1/500 → a = 200/500 = 0.4 g
0.4 g = 400 mg

Q5 · Page 61 **2 mL** — 1 in 200 = 1 mL in 200 mL
a = 400/200 = 2 mL

Q6 · Page 61 **5000 mg** — 1 in 5 = 1 g in 5 g
a = 25/5 = 5 g
5 g = 5000 mg

Q7 · Page 61 **3.1 g** — 1 in 80 = 1 g in 80 mL
a = 250/80 = 3.125 g ≈ 3.1 g

2.4 Percentage Concentration (w/v, w/w, v/v)

Slide 11

a. **3 g in 100 ml** — w/v = weight in volume → 3% w/v means 3 g in 100 ml
b. **3 g in 100 g** — w/w = weight in weight → 3% w/w means 3 g in 100 g
c. **3 ml in 100 ml** — v/v = volume in volume → 3% v/v means 3 ml in 100 ml

Slide 12 **20 g in 500 ml** — a) Convert percentage to number of grams: 4% = 4 g in 100 ml
b) Cross multiply: 4 g → 100 ml, x → 500 ml
x = 4 × 500 / 100 = 20 g in 500 ml
OR use the formula: Percentage / 100 × Number = 4 / 100 × 500 = 20 g

Slide 13 **1.33 ml** — a) Work out what 0.3% means: 0.3 g in 100 ml
b) Convert 0.3 g to mg: 0.3 × 1000 = 300 mg
c) Cross multiply: 300 mg → 100 ml, 4 mg → x
x = 4 × 100 / 300 = 1.33 ml
Formula: Amount you want / amount you got × volume = 4/300 × 100 = 1.33 ml

Example 4.15 · Page 59 **150 mg** — 0.03% w/v = 0.03 g in 100 mL = 30 mg in 100 mL

aluminium acetate (mg)	x	30
product (mL)	500	100

By spotting: 100 × 5 = 500, so x = 30 × 5 = 150 mg

Q9 · Page 62 **400 mg** — 8% = 8 g in 100 mL
8/100 × 5 = 0.4 g = 400 mg

Q10 · Page 62 **18 mL** — 3% = 3 mL in 100 mL
3/100 × 600 = 18 mL

Q11 · Page 62 **6800 mg** — 3.4% = 3.4 g in 100 mL
3.4/100 × 200 = 6.8 g = 6800 mg

Q12 · Page 62 **10 mg** — 0.01% = 0.01 g in 100 mL
0.01 g = 10 mg

Q13 · Page 62 **7.5 g** — 3% w/w = 3 g in 100 g
3/100 × 250 = 7.5 g

Q8 · Page 62 **4 mL** — 2.5% = 2.5 g in 100 mL
x mL contains 0.1 g: x = 0.1 × 100 / 2.5 = 4 mL

Q14 · Page 62 **8.8 mL** — 17% = 17 g in 100 mL
x = 1.5 × 100 / 17 = 8.82 ≈ 8.8 mL

Q15 · Page 62 **25 mL** — 20% = 20 g in 100 mL
x = 5 × 100 / 20 = 25 mL

Q16 · Page 62 **46.7 mL** — 15% = 15 g in 100 mL
x = 7 × 100 / 15 = 46.67 ≈ 46.7 mL

Q17 · Page 62 **441.2 mL** — 34% = 34 g in 100 mL
x = 150 × 100 / 34 = 441.18 ≈ 441.2 mL

Example 4.6 · Page 52 **12% w/w** — Percentage concentration = grams in 100 g
12 g in 100 g = 12% w/w

Example 4.8 · Page 53 **0.9% w/v** — Convert mg to g: 900 mg = 0.9 g
0.9 g in 100 mL = 0.9% w/v

Q3 · Page 61
(a) 2% — 100 mg = 0.1 g
0.1/5 × 100 = 2%
(b) 3% — 0.15/5 × 100 = 3%
(c) 4% — 0.2/5 × 100 = 4%
(d) 6% — 0.3/5 × 100 = 6%
(e) 10% — 0.5/5 × 100 = 10%

Example 4.7 · Page 52 **0.2% w/v** —

potassium permanganate (g)	1	x
product (mL)	500	100

Divide 500 by 5 to get 100, so divide 1 by 5: x = 1/5 = 0.2
Percentage = 0.2% w/v

Example 4.9 · Page 53 **1% w/v** — 10 mg = 0.01 g → 0.01 g in 1 mL
In 100 mL: 0.01 × 100 = 1 g
Percentage = 1% w/v

2.5 mg/ml Concentrations & Amount Strength

Slide 14 **9 mg/ml** — Sodium chloride 0.9% = 0.9 g in 100 ml
= 900 mg in 100 ml
= 9 mg in 1 ml (9 mg/ml)

Slide 16
a. 9 mg/ml — 0.9 × 10 = 9 mg/ml
b. 0.9% — 9 / 10 = 0.9%

Example 4.1 · Page 47 **9 mg/mL = 0.009 g/mL** —

sodium chloride (mg)	900	z
water (mL) to	100	1

By spotting: z = 900 ÷ 100 = 9 → 9 mg/mL
9 mg = 0.009 g → 0.009 g/mL

Example 4.12 · Page 56 **0.05% w/v; 0.5 mg/mL** — 1 in 2000 w/v = 1 g in 2000 mL

	ratio	percentage	amount (g/mL)
solid (g)	1	p	a
product (mL)	2000	100	1

1/2000 = p/100 → p = 100/2000 = 0.05
1/2000 = a/1 → a = 0.0005 g = 0.5 mg
0.05% w/v or 0.5 mg/mL

2.6 Parts per Million (ppm)

Example 4.4 · Page 50 **0.7 mg/L** — 0.7 ppm = 0.7 g in 1,000,000 mL
0.7 g = 700 mg; 1,000,000 mL = 1000 L
700 mg per 1000 L = 0.7 mg/L

Example 4.5 · Page 51 **4 L** — 0.25 ppm = 0.25 g in 1,000,000 mL = 250 mg in 1,000,000 mL

fluoride (mg)	250	1
product (mL)	1000 000	y

y = 1,000,000 / 250 = 4000 mL = 4 L

2.7 Converting Between Expressions of Concentration

Example 4.10 · Page 54 **1 in 25 v/v; 4% v/v** —

	volume	ratio	percentage
ethanol (mL)	20	1	p
product (mL)	500	r	100

500/20 = r/1 → r = 25
p = 20/5 = 4 (500 ÷ 5 = 100)
1 in 25 v/v or 4% v/v

Example 4.11 · Page 55 **2.5% w/w; 0.025 g/g** —

	ratio	percentage	amount (g/g)
ingredient (g)	1	p	a
product (g)	40	100	1

1/40 = p/100 → p = 100/40 = 2.5
1/40 = a/1 → a = 0.025
2.5% w/w or 0.025 g/g

Q1 · Page 60

(a) 4% — 1/25 × 100 = 4%

(b) 5% — 100/20 = 5%

(c) 2% — 100/50 = 2%

(d) 0.125% — 100/800 = 0.125%

(e) 0.2% — 100/500 = 0.2%

(f) 0.05% — 100/2000 = 0.05%

(g) 0.33% — 100/300 = 0.333… ≈ 0.33%

Example 4.13 · Page 57

1 in 20 v/v; 0.05 mL/mL — 5% v/v = 5 mL in 100 mL

	percentage	ratio	amount (mL/mL)
ingredient (mL)	5	1	a
product (mL)	100	r	1

r = 100/5 = 20

a = 5/100 = 0.05

1 in 20 v/v; 0.05 mL/mL

Example 4.14 · Page 57

10% w/w; 1 in 10 w/w; 0.1 g/g — For w/w the product = ingredient + vehicle = 5 + 45 = 50 g

p = 5/50 × 100 = 10

r = 50/5 × 1 = 10

a = 5/50 × 1 = 0.1

10% w/w or 1 in 10 w/w or 0.1 g/g

Q2 · Page 61

(a) 20%, 1 in 5 — 25/125 × 100 = 20%

125/25 = 5 → 1 in 5

(b) 40%, 1 in 2.5 — 50/125 × 100 = 40%

125/50 = 2.5 → 1 in 2.5

(c) 48%, 1 in 2.1 — 60/125 × 100 = 48%

125/60 = 2.08 → 1 in 2.1

(d) 4%, 1 in 25 — 5/125 × 100 = 4%

125/5 = 25 → 1 in 25

(e) 5.6%, 1 in 17.9 — 7/125 × 100 = 5.6%

125/7 = 17.86 → 1 in 17.9

2.8 Heparin & Insulin (Units)

Slide 18

1 ml of 10000 units/ml + 0.6 ml of 5000 units/ml — Total dose: 13000 units

You will need: 1 × 10000 units/ml (1 ml ampule) plus 1 × 5000 units/ml (1 ml ampule)

You need to withdraw 3000 units from the 5000 units ampule:

3000 / 5000 × 1 = 0.6 ml

So, you take 1 ml of the 10000 units/ml ampule + 0.6 ml of the 5000 units/ml ampule

---

2.9 Practice Calculations (Lecture)

Slide 20 **9 g** — 0.9% = 0.9 g in 100 ml
1 litre = 1000 ml
0.9 g → 100 ml, x → 1000 ml
x = 0.9 × 1000 / 100 = 9 g

Slide 21 **30 ml (3 ampules)** — 10% = 10 g in 100 ml = 1 g in 10 ml (one ampule = 1 g)
1 g → 10 ml, 3 g → x
x = 3 × 10 / 1 = 30 ml
So withdraw 30 ml = 3 × 10 ml ampules

Slide 22 **10% w/w; 1 in 10; 0.1 g/g** — Total product = 5 + 45 = 50 g
Percentage = 5/50 × 100 = 10% w/w
Ratio = 50/5 = 10 → 1 in 10
Amount strength = 5/50 = 0.1 g/g

3 Dilutions

In this lecture

1. Simple Dilutions 2 lecture · 2 textbook
2. Stock Solutions & Serial Dilutions 3 lecture · 9 textbook
3. Concentrated Waters 1 lecture · 1 textbook
4. Multiple Dilutions 1 lecture · 12 textbook
5. Triturations 1 lecture · 1 textbook
6. Powder Calculations 1 lecture · 3 textbook
7. Mixing Concentrations 2 lecture · 7 textbook

3.1 Simple Dilutions

Lecture Problems

Slides 4-5 You need to prepare 500 ml of 45% solution. What volume of stock solution do you need which when diluted to 500 ml will give 45% solution?

Volume of stock = ? and volume of water = ?

Slides 6-7 *Student activity:* 300 ml of 1 in 25 w/v solution diluted to 800 ml. What is the concentration of the diluted product in % and mg/ml?

a. % w/v = ?

---

b. mg/ml = ?

---

c. Ratio strength = ?

Textbook Problems

Example 5.2 · Page 67 100 mL of a 1 in 50 w/v solution is diluted to 1000 mL. Find the concentration of the diluted product as a percentage strength, a ratio strength and an amount strength expressed as mg/mL.

% , ratio and mg/mL = ?

Example 5.1 · Page 66 A product consists of a solid ingredient in a solid vehicle; 2 g of the solid ingredient is contained in 500 g of product. A further 250 g of vehicle is added to the product. Find the concentration of the diluted mixture as a percentage strength, a ratio strength and an amount strength in mg/g.

% , ratio and mg/g = ?

3.2 Stock Solutions & Serial Dilutions

Lecture Problems

Slides 8-9 You need to prepare 1 litre of 20% solution from 80% stock solution.

Stock volume = ? and diluent = ?

Slide 10 You need to prepare one litre of potassium chloride solution 1 in 10,000. You have a stock solution 10 g/L. How much of the stock solution do you need?

Stock volume = ?

Slide 13 To what volume must 350 ml of 40% w/v solution be diluted to produce a 12% solution?

Final volume = ?

Textbook Problems

Example 5.4 · Page 69 What volume of a 40% v/v solution needs to be used to produce 500 mL of 5% v/v solution?

Volume = ?

Q10 · Page 93 How many millilitres of 90% alcohol when diluted to 135 mL produces 60% alcohol? (Assume no contraction in volume.)

Volume = ?

Example 5.3 · Page 68 What volume of a 1 in 400 v/v solution is needed to produce 5 L of a 1 in 2000 v/v solution?

Volume = ?

Q12 · Page 94 What volume of a 0.6% w/v solution is required to produce 200 mL of 1 in 10 000 solution?

Volume = ?

Q13 · Page 94 What volume of a 1 in 50 solution is required to produce 300 mL of a 0.3% w/v solution?

Volume = ?

Example 5.5 · Page 70 To what volume must 250 mL of a 25% w/v solution be diluted to produce a 10% solution?

Final volume = ?

Q6 · Page 93 How many millilitres of water must be added to 180 mL of 40% v/v solution in order to produce a 5% v/v solution?

Water = ?

Q9 · Page 93 What volume of 100 mg/5 mL solution is required to be diluted to give 200 mL of 75 mg/5 mL solution?

Volume = ?

Q16 · Page 94 Calculate the volume of 0.2% w/v potassium permanganate solution required to produce 1000 mL of a 100 ppm solution.

Volume = ?

3.3 Concentrated Waters

Lecture Problems

Slide 11 Concentrated waters (rose water, peppermint water, chloroform water) are used to produce single-strength products. They are intended for dilution in a ratio of 1 in 40.

What does "1 in 40" mean here?

Textbook Problems

Worked example · Page 72 Make volumes of single-strength chloroform water (50 mL, 100 mL, 200 mL, 300 mL, 500 mL) from chloroform water concentrate (1 : 40).

a. Concentrate (mL) for each volume = ?

---

b. Double-strength chloroform water — concentrate for the same volumes = ?

3.4 Multiple Dilutions

Lecture Problems

Slide 12 *Student activity:* What weight of ingredient is required to produce 500 ml of a solution, when 3 ml diluted to 40 ml, produces a strength of 0.5% w/v solution?

Weight = ?

Textbook Problems

Example 5.10 · Page 81 What weight of ingredient is required to produce 1000 mL of a solution such that, when 2.5 mL of it is diluted to 50 mL with water, it gives a 0.25% w/v solution?

Weight = ?

Example 5.12 · Page 83 What weight of ingredient is required to produce 250 mL of a solution such that 10 mL of this solution diluted to 150 mL gives a 0.1% w/v solution?

Weight = ?

Q1 · Page 93 What weight of potassium permanganate is required to produce 300 mL of solution such that 5 mL of this solution diluted to 250 mL gives a 0.01% w/v solution?

Weight = ?

Q2 · Page 93 What weight of potassium permanganate is required to produce 500 mL of solution such that 25 mL of this solution diluted to 500 mL gives a 0.025% w/v solution?

Weight = ?

Q3 · Page 93 What weight of zinc sulfate is required to produce 150 mL of solution such that 10 mL of this solution diluted to 200 mL gives a 1% w/v solution?

Weight = ?

Q8 · Page 93 What weight of potassium permanganate is needed to produce 300 mL of solution such that if 1 mL is diluted to 20 mL a 0.1% w/v solution is produced?

Weight = ?

Example 5.11 · Page 82 What weight of malachite green oxalate is required to produce 500 mL of solution such that 25 mL of this solution diluted to 4000 L gives a 1 in 2 000 000 solution?

Weight = ?

Q4 · Page 93 What weight of potassium permanganate is required to produce 200 mL of solution such that 50 mL of this solution diluted to 200 mL will produce a 1 in 400 solution?

Weight = ?

Q7 · Page 93 What weight of copper sulfate is required to produce 150 mL of solution such that 30 mL of this solution diluted to 150 mL produces a 1 in 6000 solution?

Weight = ?

Example 5.13 · Page 84 How many millilitres of 0.5% w/v solution are required to produce 250 mL of 1 in 5000 solution?

Volume = ?

Example 5.14 · Page 85 How many millilitres of a 1 in 80 w/v solution are required to make 500 mL of a 0.02% solution?

Volume = ?

Q5 · Page 93 What weight of malachite green oxalate is required to produce 300 mL of a solution such that 25 mL of this solution diluted to 2000 L gives a 0.5 ppm solution?

Weight = ?

3.5 Triturations

Lecture Problems

Slide 14 You need to prepare 100 ml of the following formula:

Hyoscine hydrobromide	500 mcg
Water to	5 ml

a. Amount of hyoscine in 100 ml = ?

---

b. Minimum weight on the scale is 100 mg. How do you prepare it?

Textbook Problems

Example 5.6 · Page 73 How would you prepare 100 mL of a preparation to the following formula?

hyoscine hydrobromide (micrograms)	500
chloroform water (mL) to	5

(Same as the lecture question on Slide 14.)

How do you prepare it?

3.6 Powder Calculations

Lecture Problems

Slide 15 Powders must weigh a minimum of 120 mg. Prepare seven powders each containing 80 mg of drug x.

Drug, diluent and total weight for 1 and 7 powders = ?

Textbook Problems

Example 5.7 · Page 76 Prepare five powders each containing 100 mg of paracetamol.

Weights = ?

Example 5.8 · Page 76 Prepare five powders each containing 10 mg of hyoscine hydrobromide.

How do you prepare them?

Example 5.9 · Page 78 Prepare six powders each containing 0.5 mg of drug.

How do you prepare them?

3.7 Mixing Concentrations

Lecture Problems

Slide 16 What is the final strength (% v/v) if 200 ml of 40% v/v solution is added to 300 ml of 70% v/v solution?

Final % v/v = ?

Slide 17 What proportion of 60% v/v solution and 10% v/v solution would produce a 500 ml of 40% v/v solution mixture?

V1 and V2 = ?

Textbook Problems

Example 5.15 · Page 86 What is the final % v/v of a solution if 200 mL of 40% v/v solution is added to 300 mL of 70% v/v solution? (Same as the lecture question on Slide 16.)

Final % v/v = ?

Q11 · Page 94 When 70 mL of sodium chloride 0.9% w/v solution is combined with 100 mL of 1.7% w/v sodium chloride solution what is the final strength of the solution?

Final strength = ?

Q14 · Page 94 Calculate the volumes of two stock solutions of 100 mg/5 mL and 25 mg/5 mL that must be mixed to produce 500 mL of 75 mg/5 mL solution.

Volumes = ?

Q15 · Page 94 What volumes of two stock solutions of 30% w/v and 60% w/v are required to make 300 mL of 40% w/v solution?

Volumes = ?

Example 5.16 · Page 88 A suspension used to be provided as two strengths, both 100 mg/5 mL and 25 mg/5 mL solutions, which could be mixed to produce intermediate doses. If we want to produce a product that has a strength of 75 mg/5 mL, what proportion of the two mixtures do we need?

Proportion = ?

Example 5.17 · Page 90 Find the proportions of the same two suspensions given in Example 5.16 that would be required to produce a 50 mg/5 mL suspension.

Proportion = ?

Example 5.18 · Page 91 What proportions of 90% v/v and 50% v/v ethanol mixtures would produce a 70% v/v mixture? Assume no contraction.

Proportion = ?

Answers · Lecture 3

3.1 Simple Dilutions

Slides 4-5 **225 ml stock + 275 ml water** — 1) 45% v/v means 45 ml in 100 ml
2) 45 → 100
x → 500 x = 45 × 500 / 100 = 225 ml
3) Dilution: 500 − 225 = 275 ml of water
Formula (Slide 5): stock (ml) = % of final solution / 100 × final volume = 45/100 × 500 = 225 ml

Slides 6-7 **a. 1.5% w/v** — 1 in 25 → 1 g in 25 ml
1 g → 25 ml
x → 300 ml x = 12 g
After dilution the 12 g stays the same, the volume becomes 800 ml:
12 g → 800 ml
x → 100 ml x = 1.5 g → 1.5%

b. 15 mg/ml — 1.5 g → 100 ml
x → 1 ml x = 0.015 g = 15 mg → 15 mg/ml

c. 1 in 66.7 (≈ 1:67) — 1 → x
1.5 g → 100 ml
x = 100 / 1.5 = 66.6 → 1 : 67

Example 5.2 · Page 67 **0.2% w/v; 1 in 500 w/v; 2 mg/mL** — 1 in 50 = 1 g in 50 mL

ingredient (g)	1	g
product (mL)	50	100

g = 2 g in 100 mL
After dilution: 2 g in 1000 mL

	amount	percentage	ratio	g/mL
ingredient (g)	2	p	1	a
product (mL)	1000	100	r	1

p = 0.2
r = 1000/2 = 500
a = 0.002 g = 2 mg
→ 0.2% w/v, 1 in 500 w/v, 2 mg/mL

Example 5.1 · Page 66 **0.27% w/w; 1 in 375 w/w; 2.7 mg/g** — Ingredient stays 2 g; product = 500 + 250 = 750 g

	amount	percentage	ratio	g/g
ingredient (g)	2	p	1	a
product (g)	750	100	r	1

p = 2 × 100 / 750 = 0.27
r = 750 / 2 = 375
a = 2 / 750 = 0.0027 g = 2.7 mg
→ 0.27% w/w, 1 in 375 w/w, 2.7 mg/g

3.2 Stock Solutions & Serial Dilutions

Slides 8–9 **250 ml stock + 750 ml water** — 1) 20% → 20 ml in 100 ml
2) Units needed for 1 litre: 20 × 1000 / 100 = 200 ml
3) Stock containing 200 units: 200 × 100 / 80 = 250 ml
4) Dilution: 1000 − 250 = 750 ml water
Formula (Slide 9): stock = % final / % stock × final volume = 20/80 × 1000 = 250 ml

Slide 10 **10 ml** — 1) 1 in 10,000 → 1 g in 10,000 ml
2) 1 g → 10000 ml
x → 1000 ml x = 0.1 g = 100 mg
3) 10 g → 1000 ml
0.1 g → x x = 10 ml
With the formula: 10 g/L = 1%, 1 in 10,000 = 0.01% → 0.01/1 × 1000 = 10 ml

Slide 13 **1166 ml (1166.7 ml)** — x g → 350 ml
40 g → 100 ml x = 140 g
140 g → x ml
12 g → 100 ml x = 140 × 100 / 12 = 1166.7 ml

Example 5.4 · Page 69 **62.5 mL** —

ingredient (mL)	5	x
product (mL)	100	500

x = 25 mL

ingredient (mL)	40	25
product (mL)	100	y

y = 100 × 25 / 40 = 62.5 mL

Q10 · Page 93 **90 mL** — Alcohol in final: 135 × 60/100 = 81 mL
90 mL → 100 mL
81 mL → x x = 81 × 100 / 90 = 90 mL

Example 5.3 · Page 68 **1000 mL (1 L)** — 5 L = 5000 mL

ingredient (mL)	1	x
product (mL)	2000	5000

x = 5000/2000 = 2.5 mL

ingredient (mL)	1	2.5
product (mL)	400	y

y = 400 × 2.5 = 1000 mL
1 L of the 1 in 400 is diluted to 5 L.

Q12 · Page 94 **3.3 mL** — Ingredient: 1 g → 10 000 mL, x → 200 mL → x = 0.02 g
0.6 g → 100 mL
0.02 g → y y = 0.02 × 100 / 0.6 = 3.33 mL

Q13 · Page 94 **45 mL** — Ingredient: 300 × 0.3/100 = 0.9 g
1 g → 50 mL
0.9 g → y y = 45 mL

Example 5.5 · Page 70 **625 mL** —

ingredient (g)	25	x
product (mL)	100	250

x = 25 × 250 / 100 = 62.5 g

ingredient (g)	10	62.5
product (mL)	100	y

y = 625 mL

Q6 · Page 93 **1260 mL** — Ingredient: 180 × 40/100 = 72 mL
72 mL → 5% → final volume = 72 × 100 / 5 = 1440 mL
Water = 1440 − 180 = 1260 mL

Q9 · Page 93 **150 mL** — Drug needed: 75 mg/5 mL × 200 mL = 3000 mg
100 mg → 5 mL
3000 mg → x x = 3000 × 5 / 100 = 150 mL

Q16 · Page 94 **50 mL** — 100 ppm = 100 g in 1 000 000 mL → in 1000 mL: 0.1 g
0.2 g → 100 mL
0.1 g → y y = 50 mL

3.3 Concentrated Waters

Slide 11 **1 part concentrated water + 39 parts water** — 1 part of concentrate is made up to 40 parts in total with water → 1 + 39 = 40

Worked example · Page 72 **a. 1.25, 2.5, 5, 7.5, 12.5 mL —**

concentrate (mL)	1	a	b	c	d	e
water (mL) to	40	50	100	200	300	500

c = 200/40 = 5

b = c/2 = 2.5, a = b/2 = 1.25, d = b × 3 = 7.5, e = b × 5 = 12.5

b. 2.5, 5, 10, 15, 25 mL (page 73) — Double strength → twice the concentrate

concentrate (mL)	2	2.5	5	10	15	25
water (mL) to	40	50	100	200	300	500

3.4 Multiple Dilutions

Slide 12 **33.33 g** — Final: 0.5 g → 100 ml

y → 40 ml y = 0.2 g

These 0.2 g came from the 3 ml taken:

0.2 g → 3 ml

z → 500 ml z = 0.2 × 500 / 3 = 33.33 g

Example 5.10 · Page 81 **50 g** —

ingredient (g)	0.25	y
water (mL) to	100	50

y = 0.125 g (in the 2.5 mL)

ingredient (g)	z	0.125
water (mL) to	1000	2.5

z = 0.125 × 1000 / 2.5 = 50 g

Example 5.12 · Page 83 **3.75 g** —

ingredient (g)	y	0.1
product (mL)	150	100

y = 150 × 0.1 / 100 = 0.15 g

ingredient (g)	z	0.15
product (mL)	250	10

z = 0.15 × 250 / 10 = 3.75 g

Q1 · Page 93 **1.5 g** — 250 mL of 0.01%: 0.01 × 250 / 100 = 0.025 g (in the 5 mL)

0.025 g → 5 mL

z → 300 mL z = 0.025 × 300 / 5 = 1.5 g

Q2 · Page 93 **2.5 g** — 500 mL of 0.025%: 0.025 × 500 / 100 = 0.125 g (in the 25 mL)
0.125 g → 25 mL
z → 500 mL z = 2.5 g

Q3 · Page 93 **30 g** — 200 mL of 1%: 2 g (in the 10 mL)
2 g → 10 mL
z → 150 mL z = 30 g

Q8 · Page 93 **6 g** — 20 mL of 0.1%: 0.02 g (in the 1 mL)
0.02 g → 1 mL
z → 300 mL z = 6 g

Example 5.11 · Page 82 **40 g** — 4000 L = 4 000 000 mL

malachite green (g)	1	y
water (mL) to	2 000 000	4 000 000

y = 2 g (all in the 25 mL)

malachite green (g)	z	2
water (mL) to	500	25

z = 2 × 500 / 25 = 40 g

Q4 · Page 93 **2 g** — 1 g → 400 mL, x → 200 mL → x = 0.5 g (in the 50 mL)
0.5 g → 50 mL
z → 200 mL z = 2 g

Q7 · Page 93 **125 mg** — 1 g → 6000 mL, x → 150 mL → x = 0.025 g (in the 30 mL)
0.025 g → 30 mL
z → 150 mL z = 0.125 g = 125 mg

Example 5.13 · Page 84 **10 mL** —

ingredient (g)	1	x
product (mL)	5000	250

x = 250/5000 = 0.05 g

ingredient (g)	0.5	0.05
product (mL)	100	y

y = 10 mL

Example 5.14 · Page 85 **8 mL** —

ingredient (g)	0.02	x
product (mL)	100	500

x = 0.1 g

ingredient (g)	1	0.1
product (mL)	80	y

y = 8 mL

Q5 · Page 93 **12 g** — 2000 L = 2 000 000 mL
0.5 ppm = 0.5 g in 1 000 000 mL → in 2 000 000 mL: 1 g (in the 25 mL)
1 g → 25 mL
z → 300 mL z = 12 g

3.5 Triturations

Slide 14 **a. 10 mg** — 500 mcg → 5 ml
x mcg → 100 ml x = 10 000 mcg = 10 mg

b. 100 mg in 1000 ml — or 100 mg in 50 ml, take 5 ml + 95 ml water — Method 1: 10 mg → 100 ml
100 mg → x x = 1000 ml
Method 2: dissolve 100 mg in 50 ml
100 mg → 50 ml
10 mg → x x = 5 ml
5 ml + 95 ml H_2O = 100 ml

Example 5.6 · Page 73 **Weigh 100 mg, make up to 10 mL, take 1 mL and dilute to 100 mL** —

hyoscine (micrograms)	500	y
chloroform water (mL) to	5	100

y = 10 000 micrograms = 10 mg — can't weigh < 100 mg
Option 1: 100 mg made up to 1000 mL (x = 1000)
Alternative: 100 mg made up to 10 mL → 10 mg is in z = 1 mL → dilute 1 mL to 100 mL (only 110 mL used)
Or: 100 mg in 100 mL → take a = 10 mL → dilute to 100 mL (200 mL used)

3.6 Powder Calculations

Slide 15 **Drug 560 mg + diluent 280 mg = 840 mg total** —

	1 powder	7 powders
drug (mg)	80	560
diluent (mg)	X1 = 40	X2 = 280
total (mg)	120	840

X1 = 120 − 80 = 40 mg; then × 7

Example 5.7 · Page 76 **500 mg paracetamol + 100 mg diluent → 5 × 120 mg** — y = 120 − 100 = 20 mg diluent per powder

powders	1	5
paracetamol (mg)	100	500
diluent (mg)	20	100
total (mg)	120	600

Example 5.8 · Page 76 **Triturate: 100 mg drug + 900 mg diluent; take 500 mg + 100 mg diluent → 5 × 120 mg** — y = 120 − 10 = 110

powders	1	5
hyoscine (mg)	10	50
diluent (mg)	110	550
total (mg)	120	600

50 mg can't be weighed (< 100 mg).
Trituration: 100 mg hyoscine + 900 mg diluent = 1000 mg

hyoscine (mg)	100	50
diluent (mg)	900	m
powder (mg)	1000	n

n = 500 mg
5 × 120 = 600 → 600 − 500 = 100 mg extra diluent

Example 5.9 · Page 78 **Two triturations: take 300 mg of T2 + 420 mg diluent → 6 × 120 mg** — y = 120 − 0.5 = 119.5

powders	1	6
drug (mg)	0.5	3
diluent (mg)	119.5	717
total (mg)	120	720

T1: 100 mg drug + 900 mg diluent → 3 mg drug in 30 mg (too small to weigh)
T2: 100 mg of T1 (10 mg drug) + 900 mg diluent → 3 mg drug in z = 300 mg
6 × 120 = 720 → 720 − 300 = 420 mg diluent

3.7 Mixing Concentrations

Slide 16 **58% v/v** — V1: 40 ml → 100 ml, x → 200 ml → x = 80 ml
V2: 70 ml → 100 ml, x → 300 ml → x = 210 ml
80 + 210 = 290 ml in 500 ml
290 → 500, x → 100 → x = 58% v/v

Slide 17 **V1 = 300 ml (60%), V2 = 200 ml (10%)** — V1/V2 = (C3 − C2) / (C1 − C3)
C1 = 60, C2 = 10, C3 = 40
= (40 − 10)/(60 − 40) = 30/20 = 3/2
V1 = 3/5 × 500 = 300 ml
V2 = 2/5 × 500 = 200 ml

Example 5.15 · Page 86 **58% v/v** —

ingredient (mL)	40	x
product (mL)	100	200

x = 80

ingredient (mL)	70	y
product (mL)	100	300

y = 210
80 + 210 = 290 mL

ingredient (mL)	290	p
product (mL)	500	100

p = 58

Q11 · Page 94 **1.37% w/v** — 70 × 0.9/100 = 0.63 g
100 × 1.7/100 = 1.7 g
Total 2.33 g in 170 mL
2.33 × 100 / 170 = 1.37%

Q14 · Page 94 **333 mL of 100 mg/5 mL + 167 mL of 25 mg/5 mL** — v1/v2 = (15 − 5)/(20 − 15) = 2/1
v1 = 2/3 × 500 = 333 mL
v2 = 1/3 × 500 = 167 mL

Q15 · Page 94 **100 mL of 60% + 200 mL of 30%** — C1 = 60, C2 = 30, C3 = 40
V1/V2 = (40 − 30)/(60 − 40) = 10/20 = 1/2
V1 (60%) = 1/3 × 300 = 100 mL
V2 (30%) = 2/3 × 300 = 200 mL

Example 5.16 · Page 88

2 parts 100 mg/5 mL : 1 part 25 mg/5 mL — c1 = 100/5 = 20, c2 = 25/5 = 5, c3 = 75/5 = 15

v1/v2 = (15 − 5)/(20 − 15) = 10/5 = 2/1

Check (300 mL): 200 mL × 20 = 4000 mg + 100 mL × 5 = 500 mg → 4500 mg / 300 mL = 75 mg/5 mL ✓

Example 5.17 · Page 90

1 part 100 mg/5 mL : 2 parts 25 mg/5 mL — c1 = 20, c2 = 5, c3 = 10

v1/v2 = (10 − 5)/(20 − 10) = 5/10 = 1/2

Example 5.18 · Page 91

1 : 1 — v1/v2 = (70 − 50)/(90 − 70) = 20/20 = 1/1

Check: 100 mL + 100 mL → 90 + 50 = 140 mL ethanol in 200 mL = 70% ✓

4 Moles

In this lecture

1. Moles & Molecular Mass 3 lecture · 2 textbook
2. Amount of an Ion in a Compound 1 lecture · 8 textbook
3. Drug Salts, Bases & Hydrates 1 lecture · 15 textbook
4. Legal Classification Using Molecular Weights 3 textbook
5. Moles & Millimoles 1 lecture · 5 textbook
6. Millimoles in Infusions & Molar Solutions 4 lecture · 7 textbook

4.1 Moles & Molecular Mass

Lecture Problems

Slides 3-4 One mole = the relative atomic mass in grams. One mole of any element contains 6×10^{23} atoms (Avogadro's constant). Atomic masses: Ca = 40, Cl = 35.5, Na = 23, C = 12, H = 1.

a. How many moles are in 24 g of carbon?

b. How many moles (and atoms) are in 1 g of carbon?

Slide 5 Sodium chloride (NaCl) is made up of sodium (Na) and chloride (Cl): $NaCl \rightarrow Na^{+} + Cl^{-}$.

Molecular mass of NaCl = ? and what does 1 mole of NaCl give?

Slide 6 CO_2: C = 12, O = 16.

Molecular mass of CO_2 = ?

Textbook Problems

Worked examples · Page 166 Calculate the molecular weights (K = 39, Br = 79.9, Al = 27, Cl = 35.5, Zn = 65.4, S = 32, O = 16, H = 1, C = 12):

a. Potassium bromide, KBr

b. Aluminium chloride, $AlCl_3$

c. Zinc sulfate monohydrate, $ZnSO_4.H_2O$

d. Aspirin, $C_9H_8O_4$

Q1 · Page 189 For the following drugs calculate the molecular weight and the weight of salt containing 1 millimole of the metal ion:

a. (a) potassium chloride, KCl

b. (b) sodium chloride, NaCl

c. (c) calcium chloride, $CaCl_2.2H_2O$

d. (d) calcium lactate, $C_6H_{10}CaO_6.5H_2O$

e. (e) potassium bicarbonate, $KHCO_3$

f. (f) magnesium chloride, $MgCl_2.6H_2O$

g. (g) sodium citrate, $C_6H_5Na_3O_7.2H_2O$

h. (h) ammonium chloride, NH_4Cl

4.2 Amount of an Ion in a Compound

Lecture Problems

Slide 13 How many mg of sodium ions are contained in 850 mg tablet of sodium chloride?

Na^+ (mg) = ?

Textbook Problems

Example 10.1 · Page 167 How many milligrams of sodium ions are contained in a 600-mg tablet of sodium chloride?

Na^+ (mg) = ?

Q8 · Page 190 Calcium acetate, $C_4H_6CaO_4$.

a. Molecular weight = ?

---

b. Calcium provided by 300 mg of calcium acetate = ?

---

c. mg of calcium acetate in 1 millimole = ?

Q11 · Page 190 Sodium Fluoride BP drops contain 275 micrograms of sodium fluoride per drop. Calculate the quantity of fluoride ion provided by five drops (Na = 23, F = 19).

Fluoride = ?

Example 10.2 · Page 168 What is the percentage of lithium contained in a 400-mg tablet of lithium carbonate?

% Li = ?

Q2 · Page 189 Lithium citrate, $C_6H_5Li_3O_7.H_2O$ (Li = 7).

a. Molecular weight = ?

---

b. % of Li^+ = ?

Q3 · Page 190 The empirical formula of ferrous fumarate is $C_4H_2FeO_4$.

a. Molecular weight = ?

b. Weight of ferrous fumarate that gives 100 mg of iron = ?

Q4 · Page 190 A syrup contains ferrous fumarate 140 mg/5 mL. How much iron will a patient receive per day if the daily dosage regimen is 10 mL twice a day?

Iron per day = ?

Q7 · Page 190 Calcium gluconate (molecular weight 448.4) tablets each contain 53.4 mg of calcium (Ca = 40.08).

a. Calcium gluconate in each tablet = ?

b. Millimoles of calcium per tablet = ?

4.3 Drug Salts, Bases & Hydrates

Lecture Problems

Slide 14 What weight of ferrous gluconate contains the same quantity of iron as tablet containing 300 mg of ferrous sulphate (dried)? ($FeSO_4$ = 152, ferrous gluconate = 482, Fe = 56)

Ferrous gluconate = ?

Textbook Problems

Example 10.3 · Page 169 What weight of ferrous gluconate contains the same quantity of iron as a tablet containing 200 mg of ferrous sulfate (dried)?

Ferrous gluconate = ?

Example 10.4 · Page 171 Erythromycin 734, erythromycin ethyl succinate 862, erythromycin stearate 1018. Calculate the amounts of erythromycin ethyl succinate and erythromycin stearate that are equivalent to 250 mg erythromycin.

Ethyl succinate and stearate = ?

Example 10.5 · Page 172 Adrenaline injection contains 100 micrograms of adrenaline (as the acid tartrate) per mL. MW adrenaline 183, adrenaline acid tartrate 333. Calculate the amount of adrenaline acid tartrate in each mL.

Acid tartrate per mL = ?

Q12 · Page 191 Ranitidine Injection BP contains ranitidine 25 mg/mL (as the hydrochloride). Calculate the amount of ranitidine hydrochloride needed to prepare 100 mL. (MW: ranitidine hydrochloride 350.9, ranitidine 314.4)

Ranitidine HCl = ?

Q17 · Page 191 A patient requires 300 mg of quinine (MW 324.4). What weight of quinine hydrobromide (MW 423.3) will provide 300 mg of quinine?

Quinine HBr = ?

Example 10.6 · Page 173 How many grams of salbutamol sulfate will contain the same amount of salbutamol base as 5 g of salbutamol? MW salbutamol 239, salbutamol sulfate $(C_{13}H_{21}NO_3)_2.H_2SO_4$ 577.

Salbutamol sulfate = ?

Example 10.7 · Page 174 BNF: quinine (anhydrous base) 100 mg ≡ quinine bisulfate 169 mg ≡ quinine dihydrochloride 122 mg. Quinine base MW = 324. Estimate the molecular weights of the two salts.

MW of bisulfate and dihydrochloride = ?

Example 10.8 · Page 174 Show that a 300-mg tablet of quinine bisulfate contains less quinine than a 300-mg tablet of quinine sulfate (BNF equivalent for quinine sulfate = 121).

Quinine base in each = ?

Example 10.9 · Page 176 1 L of Paediatric Ferrous Sulfate Oral Solution BP contains 12 g of ferrous sulfate. You need 150 mL, but only ferrous sulfate anhydrous is available. MW ferrous sulfate 278, anhydrous 152. How much anhydrous should be used?

Anhydrous ferrous sulfate = ?

Example 10.10 · Page 177 How much codeine phosphate sesquihydrate (MW 424) is equivalent to 30 mg of codeine phosphate (hemihydrate, MW 406)?

Sesquihydrate = ?

Q6 · Page 190 Glucose intravenous infusion contains 5% glucose expressed as anhydrous glucose.

a. Anhydrous glucose (MW 180) for 500 mL = ?

---

b. Glucose monohydrate (MW 192) for 750 mL = ?

Q14 · Page 191 A syrup contains chloroquine sulfate (MW 436), 68 mg/5-mL dose. Calculate the amount of chloroquine base (MW 319.9) in each 5-mL dose.

Base = ?

Q15 · Page 191 How much chloroquine base (MW 319.9) is equivalent to a tablet containing 250 mg chloroquine phosphate (MW 515.9)?

Base = ?

Q16 · Page 191 How much morphine (MW 285.3) is contained in 50 mL of a 2% solution of morphine tartrate (MW 774.8)?

Morphine = ?

Q18 · Page 191 How much hyoscine (MW 303.4) is contained in a 20-mg tablet of hyoscine butylbromide (MW 440.4)?

Hyoscine = ?

4.4 Legal Classification Using Molecular Weights

Textbook Problems

Example 10.11 · Page 178 Ephedrine HCl can be sold over the counter if the maximum dose ≡ 30 mg ephedrine and the maximum daily dose ≡ 60 mg ephedrine. MW ephedrine HCl 202, ephedrine 165. What are these limits as ephedrine hydrochloride?

Max dose and max daily dose = ?

Example 10.12 · Page 179 Morphine salts are CD Inv POM up to 0.2% (as anhydrous morphine base). MW base 285, morphine HCl 376. Maximum morphine hydrochloride in 200 mL?

Morphine HCl = ?

Example 10.13 · Page 180 Codeine is CD Inv POM up to 2.5% (as base) or 100 mg base per dosage unit. MW codeine base 317, codeine phosphate 406.

a. Max codeine phosphate in 100 mL syrup = ?

---

b. Max codeine phosphate per dosage unit = ?

4.5 Moles & Millimoles

Lecture Problems

Slide 7 In practice, moles are too big for everyday use, so millimoles are used.

1 millimole = ? mole

Textbook Problems

Example 10.14 · Page 182 How many millimoles each of Na^+ and HCO_3^- are there in a 500-mg sodium bicarbonate capsule? ($NaHCO_3$ = 84)

mmol = ?

Example 10.15 · Page 183 Calculate the number of moles of chloride ion in 500 mg of calcium chloride ($CaCl_2$ = 110).

Chloride = ?

Q9 · Page 190 Calcium phosphate, $Ca_3(PO_4)_2$.

a. Molecular weight = ?

---

b. A powder contains 3.3 g. Millimoles of calcium ion = ?

Example 10.16 · Page 184 Calculate the number of moles of water in 500 mg of $CaCl_2.6H_2O$ (MW 219).

Water = ?

Q13 · Page 191 "Low sodium" means less than 1 millimole of sodium per tablet or 10-mL dose. What is the maximum amount of sodium chloride in a 10-mL dose?

NaCl = ?

4.6 Millimoles in Infusions & Molar Solutions

Lecture Problems

Slides 8-9 How many millimoles of sodium are there in 500 ml infusion containing 1.8 mg/ml sodium chloride?

mmol Na = ?

Slide 10 How many millimoles of sodium are there in a 1 litre infusion containing 1800 mg/litre sodium chloride?

mmol = ?

Slide 11 How many millimoles of sodium are in one litre sodium chloride infusion 0.9%?

mmol = ?

Slide 12 *Student activity:* You need to add 35 mmol of potassium chloride to one litre of infusion. You have ampule containing potassium chloride 2 g/10 ml. Calculate how much do you need to withdraw? (KCl = 39 + 35.5 = 74.5)

Volume = ?

Textbook Problems

Example 10.19 · Page 186 How many millimoles of sodium and chloride ions are contained in 1 L of Sodium Chloride Infusion BP (0.9%)? (Same as the lecture question on Slide 11.)

mmol = ?

Q5 · Page 190 A syrup contains 7.5% of potassium chloride. How many millimoles of K^+ are contained in 1 mL?

mmol K^+ = ?

Q10 · Page 190 Magnesium sulfate injection is 50% magnesium sulfate ($MgSO_4.7H_2O$). How many millimoles of magnesium ion per mL?

mmol/mL = ?

Example 10.17 · Page 185 How many moles of solute are there in 20 mL of a 1 mol/L solution?

Moles = ?

Example 10.18 · Page 185 How much sodium chloride is required to make 25 mL of a 0.5 molar solution? How many moles of sodium ion will there be?

NaCl and Na^+ = ?

Example 10.21 · Page 188 How much sodium fluoride (MW 42) is required to make 250 mL of a mouthwash containing 0.012 mol/L?

NaF = ?

Example 10.20 · Page 187 How many millimoles/L of chloride ions are in Potassium Chloride (0.3%) and Sodium Chloride (0.9%) Infusion BP? (KCl 74.5, NaCl 58.5)

Chloride = ?

Answers · Lecture 4

4.1 Moles & Molecular Mass

Slides 3-4

a. 2 moles — 12 g of carbon = 1 mole
24 g → 24 / 12 = 2 moles

b. 1/12 mole = 6/12 × 10^{23} atoms — 1 g / 12 = 1/12 mole
atoms = 1/12 × 6 × 10^{23} = 0.5 × 10^{23}

Slide 5

58.5 g → 1 mole Na^+ + 1 mole Cl^- — Na 23 + Cl 35.5 = 58.5
One mole = the molecular mass in grams = 58.5 g
1 mole NaCl gives 1 mole of sodium and 1 mole of chloride

Slide 6

44 → 1 mole of CO_2 = 44 g — C = 12
O = 16 + 16 = 32
Total = 44
1 mole of CO_2 = 44 g (it contains 1 mole C = 12 g and 2 moles O = 32 g)

⚠️ The slide writes "one mole of C = 44 g" and "one mole of O = 44/2 = 22 g" — the correct reading is 1 mole of CO_2 = 44 g, and one mole of O atoms is always 16 g.

Worked examples · Page 166

a. 118.9 g — 39 + 79.9 = 118.9 g
b. 133.5 g — 27 + (35.5 × 3) = 133.5 g
c. 179.4 g — 65.4 + 32 + (16 × 4) + (1 × 2) + 16 = 179.4 g
d. 180 g — (9 × 12) + (8 × 1) + (4 × 16) = 180 g

Q1 · Page 189

a. MW 74.5; 74.5 mg — 39 + 35.5 = 74.5 → 1 mmol = 74.5 mg
b. MW 58.5; 58.5 mg — 23 + 35.5 = 58.5
c. MW 147; 147 mg — 40 + 71 + (2 × 18) = 147
d. MW 308; 308 mg — 72 + 10 + 40 + 96 + (5 × 18) = 308
e. MW 100; 100 mg — 39 + 1 + 12 + 48 = 100
f. MW 203; 203 mg — 24.3 + 71 + (6 × 18) = 203.3 ≈ 203
g. MW 294.1; 294.1 mg (book) — 72 + 5 + 69 + 112 + 36 = 294

⚠️ The book gives 294.1 mg (1 mmol of the salt). Because there are 3 Na per molecule, 1 mmol of Na^+ is actually in 294.1/3 ≈ 98 mg.

h. MW 53.5; 53.5 mg — 14 + 4 + 35.5 = 53.5

4.2 Amount of an Ion in a Compound

Slide 13

334 mg — 850 → x
58.5 → 23
x = 850 × 23 / 58.5 = 334 mg

Example 10.1 · Page 167 **235.9 mg** — Na + Cl = 23 + 35.5 = 58.5

	NaCl	Na
weight (mg)	600	x
molecular/atomic weight	58.5	23

x = 600 × 23 / 58.5 = 235.9 mg

Q8 · Page 190

a. **158** — 48 + 6 + 40 + 64 = 158

b. **75.9 mg** — 300 × 40 / 158 = 75.9 mg

c. **158 mg** — 1 mmol = MW in mg = 158 mg

Q11 · Page 190 **622 micrograms** — 5 × 275 = 1375 micrograms NaF

NaF = 23 + 19 = 42

1375 × 19 / 42 = 622 micrograms

Example 10.2 · Page 168 **18.92%** — Li_2CO_3 = (7 × 2) + 12 + (16 × 3) = 74 → Li = 14 parts

y = 400 × 14 / 74 = 75.7 mg

75.7 / 400 × 100 = 18.92%

Directly: 14 / 74 × 100 = 18.92%

Q2 · Page 189

a. **228** — 72 + 5 + 21 + 112 + 18 = 228

b. **9.21%** — 21 / 228 × 100 = 9.21%

Q3 · Page 190

a. **169.9** — 48 + 2 + 55.9 + 64 = 169.9 (Fe = 55.85)

b. **304 mg** — x = 100 × 169.9 / 55.85 = 304 mg

Q4 · Page 190 **184 mg** — 10 mL × 2 = 20 mL/day → 140 × 20/5 = 560 mg ferrous fumarate

Iron = 560 × 55.85 / 169.9 = 184 mg

Q7 · Page 190

a. **597.4 mg** — 53.4 × 448.4 / 40.08 = 597.4 mg

b. **1.33 mmol** — 53.4 / 40.08 = 1.33 mmol

4.3 Drug Salts, Bases & Hydrates

Slide 14 **951 mg** — Iron in 300 mg ferrous sulphate: 300 × 56 / 152 = 110.5 mg

Ferrous gluconate with 110.5 mg iron: 110.5 × 482 / 56 = 951 mg

Shortcut: 300 × 482 / 152 = 951 mg

Example 10.3 · Page 169 **634.17 mg** — Fe + S + O_4 = 56 + 32 + 64 = 152

	iron	ferrous sulfate
amount (mg)	y	200
molecular/atomic weight	56	152

y = 73.68 mg iron
Ferrous gluconate $C_{12}H_{22}FeO_{12}.2H_2O$ = 482
z = 73.68 × 482 / 56 = 634.17 mg

Example 10.4 · Page 171 **294 mg and 347 mg** — x = 250 × 862 / 734 = 294 mg
y = 250 × 1018 / 734 = 347 mg

Example 10.5 · Page 172 **182 micrograms** — z = 100 × 333 / 183 = 182 micrograms

Q12 · Page 191 **2.79 g** — 25 × 100 = 2500 mg ranitidine
2500 × 350.9 / 314.4 = 2790 mg = 2.79 g

Q17 · Page 191 **391.5 mg** — 300 × 423.3 / 324.4 = 391.5 mg

Example 10.6 · Page 173 **6.04 g** — One sulfate molecule has 2 salbutamol → equivalent MW = 239 × 2 = 478
x = 5 × 577 / 478 = 6.04 g

Example 10.7 · Page 174 **548 and 395** — x = 324 × 169 / 100 = 548
y = 324 × 122 / 100 = 395

Example 10.8 · Page 174 **Sulfate 248 mg; bisulfate 178 mg** — Sulfate: 300 × 100 / 121 = 248 mg
Bisulfate: 300 × 100 / 169 = 178 mg
178 < 248 → statement correct

Example 10.9 · Page 176 **0.984 g** — x = 12 × 150 / 1000 = 1.8 g ferrous sulfate
y = 1.8 × 152 / 278 = 0.984 g

Example 10.10 · Page 177 **31.33 mg** — z = 30 × 424 / 406 = 31.33 mg

Q6 · Page 190
a. 25 g — 5 g → 100 mL → 500 mL: 25 g
b. 40 g — Anhydrous needed: 5 × 7.5 = 37.5 g
Monohydrate: 37.5 × 192 / 180 = 40 g

Q14 · Page 191 **49.9 mg** — 68 × 319.9 / 436 = 49.9 mg

Q15 · Page 191 **155 mg** — 250 × 319.9 / 515.9 = 155 mg

Q16 · Page 191 **0.368 g (book answer)** — 2% × 50 mL = 1 g morphine tartrate
1 × 285.3 / 774.8 = 0.368 g

⚠️ Morphine tartrate contains 2 morphine molecules per molecule (like salbutamol sulfate in Example 10.6), so strictly 1 × 570.6 / 774.8 = 0.736 g. The book's answer uses 1 : 1.

Q18 · Page 191 **13.8 mg** — 20 × 303.4 / 440.4 = 13.8 mg

4.4 Legal Classification Using Molecular Weights

Example 10.11 · Page 178 **36.7 mg and 73.4 mg** — x = 30 × 202 / 165 = 36.7 mg
y = 60 × 202 / 165 = 73.4 mg

Example 10.12 · Page 179 **0.528 g** — 0.2% → 0.4 g base in 200 mL
x = 0.4 × 376 / 285 = 0.528 g

Example 10.13 · Page 180
a. **3.2 g** — 2.5 g base → 2.5 × 406 / 317 = 3.2 g
b. **128 mg** — 100 × 406 / 317 = 128 mg

4.5 Moles & Millimoles

Slide 7 **1/1000 mole** — 1 mole = 1000 mmol → 1 mmol = 0.001 mole
(1 mmol = 1000 micromoles; 1 micromole = 1000 nanomoles)

Example 10.14 · Page 182 **5.9 mmol of each** — 0.5 g / 84 = 0.0059 mole = 5.9 mmol
1 $NaHCO_3$ → 1 Na^+ + 1 HCO_3^- → 5.9 mmol each

Example 10.15 · Page 183 **9.0 mmol (0.009 mole)** — 0.5 / 110 = 0.0045 mole = 4.5 mmol $CaCl_2$
Cl_2 → 4.5 × 2 = 9.0 mmol chloride

Q9 · Page 190
a. **310** — (3 × 40) + 2 × (31 + 64) = 120 + 190 = 310
b. **32 mmol** — 3300 / 310 = 10.6 mmol salt × 3 Ca = 32 mmol

Example 10.16 · Page 184 **0.0137 mole = 13.7 mmol** — y = 0.5 × 6 / 219 = 0.0137 mole

Q13 · Page 191 **58.5 mg** — 1 mmol Na^+ = 1 mmol NaCl = 58.5 mg

4.6 Millimoles in Infusions & Molar Solutions

Slides 8-9 **15.4 mmol** — Total number = mg/ml ÷ mg of substance containing 1 mmol × volume (ml)
= 1.8 / 58.5 × 500 = 15.4 mmol
Or (Slide 9): 1.8 mg → 1 ml, x → 500 ml → 900 mg
58.5 mg → 1 mmol, 900 mg → x → 15.4 mmol

Slide 10 **30.8 mmol** — mmol/litre = mg/litre ÷ mg containing 1 mmol
= 1800 / 58.5 = 30.8 mmol

Slide 11 **154 mmol** — Total mmol = % w/v × 10 × volume ÷ mg containing 1 mmol
= 0.9 × 10 × 1000 / 58.5 = 154 mmol
Or: 0.9 g → 100 mL, x → 1000 mL → 9 g = 9000 mg
9000 / 58.5 = 154 mmol

Slide 12 **≈ 13 ml (13.04 ml)** — 1 mmol KCl = 74.5 mg → 35 mmol = 35 × 74.5 = 2607.5 mg = 2.6075 g
2 g → 10 ml
2.6075 g → x x = 2.6075 × 10 / 2 = 13.04 ml

Example 10.19 · Page 186 **154 mmol/L each** — 9 g in 1 L → 9 / 58.5 = 0.154 mole = 154 mmol
Na^+ 154 mmol and Cl^- 154 mmol

Q5 · Page 190 **1 mmol** — 7.5% → 75 mg/mL
75 / 74.5 = 1.007 ≈ 1 mmol

Q10 · Page 190 **2 mmol/mL** — MW = 24.3 + 32 + 64 + 126 = 246.3
50% → 500 mg/mL
500 / 246.3 = 2.03 ≈ 2 mmol/mL

Example 10.17 · Page 185 **0.02 mole** — y = 1 × 20 / 1000 = 0.02 mole
⚠️ The book's last line says "200 mL" by mistake — it is 20 mL.

Example 10.18 · Page 185 **0.73 g NaCl; 0.0125 mole Na^+** — y = 0.5 × 25 / 1000 = 0.0125 mole
0.0125 × 58.5 = 0.731 g
Na^+ = 0.0125 mole

Example 10.21 · Page 188 **0.126 g** — 0.012 × 42 = 0.504 g per 1 L
y = 0.504 × 250 / 1000 = 0.126 g

Example 10.20 · Page 187 **194 mmol/L** — NaCl: 9 / 58.5 = 154 mmol
KCl: 3 / 74.5 = 0.04 mole = 40 mmol
Cl^- = 154 + 40 = 194 mmol/L

5 Doses for Adults

In this lecture

1. Solid Dose Units (Tablets, Capsules...) 1 lecture · 22 textbook
2. Oral Liquid Formulations 1 lecture · 13 textbook
3. Dose Volumes Less Than 5 ml 1 lecture · 4 textbook
4. Injections & Units 6 textbook
5. Doses Based on Body Weight 3 lecture
6. Doses Based on Body Surface Area (BSA) 3 lecture

5.1 Solid Dose Units (Tablets, Capsules...)

Lecture Problems

Slide 8 A prescription requires two capsules of drug X for 5/7. How many tablet should be supplied?

Number = ?

Textbook Problems

Example 7.1 · Page 113 Calculate the total number of tablets to be dispensed if a prescription requires two tablets of drug X three times a day for 7 days.

Tablets = ?

Example 7.2 · Page 114 A prescription requires two suppositories of drug Y daily for 3/7. How many suppositories should be supplied?

Suppositories = ?

Q1 · Page 125 A patient is prescribed two tablets of drug A four times a day for 3 months. How many tablets?

Tablets = ?

Q7 · Page 125 Charcoal sachets: two sachets three times a day for 1/12. How many sachets?

Sachets = ?

Example 7.6 · Page 117 A prescription requires a patient to receive 20 mg domperidone (in tablet form) every 8 hours for 4 weeks. Domperidone is available only as 10-mg tablets. How many tablets?

Tablets = ?

Q23 · Page 127 A scored tablet contains 100 mg of drug X. Prescription: 50 mg three times a day for 10 days. How many tablets?

Tablets = ?

Q24 · Page 127 Drug Z is only available as a 62.5-mg capsule. Prescription: 125 mg four times a day for 1/12. How many capsules?

Capsules = ?

Q25 · Page 127 Drug A is available as a 500-microgram tablet. A patient requires 2 mg four times a day for 3/7 and 1.5 mg for 5/7.

a. Tablets for a 2-mg dose = ?

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b. Tablets for a 1.5-mg dose = ?

---

c. Total tablets = ?

Q26 · Page 127 Drug B: 500 mg twice a day for 5 days; available as 250-mg capsules. How many capsules?

Capsules = ?

Example 7.7 · Page 118 A patient requires 8 mg of drug X per day for 1 week followed by 7 mg per day for the next week. Drug X is available in tablets of 1 mg and 4 mg. How many tablets of each strength?

Tablets = ?

Q27 · Page 127 Drug A is available as 2.5-mg and 5-mg tablets. Prescription: 7.5 mg TDS for 5 days, then 5 mg QDS for 3 days, then 5 mg BD for 3 days, then 2.5 mg TDS for 7 days. How many of each?

Tablets = ?

Q2 · Page 125 A customer buys a pack of 100 vitamin capsules. The dosage regimen is one capsule three times a day. How many days will the pack last?

Days = ?

Q3 · Page 125 An inhaler contains 200 metered doses. If the patient inhales two doses three times a day, how long will the inhaler last?

Days = ?

Q6 · Page 125 A residential home buys a 1000 pack of senna tablets. Each of 35 residents takes two tablets each night. How long will it last?

Days = ?

Q4 · Page 125 Analgesic suppositories to be used six times a day for 5 days, then three times a day for 3 days. How many suppositories?

Suppositories = ?

Q8 · Page 125 Drug Y: three capsules twice a day for 3/7, two capsules twice a day for 3/7, one capsule twice a day for 2/52. How many capsules?

Capsules = ?

Q5 · Page 125 A patient needs 8 weeks' antimalarial treatment. Drug A: one tablet per week; drug B: two tablets daily. How many tablets of A and B?

Tablets = ?

Q11 · Page 126 One nicotine patch every 3 days. How many patches for 3 months?

Patches = ?

Q9 · Page 126 A sachet of 0.9% saline is used to bathe an eye. Bathe both eyes twice a day for 1/12. How many sachets?

Sachets = ?

Q10 · Page 126 A patient going on holiday for 3 weeks is supplied 168 tablets. Regimen: two tablets four times a day. Enough?

Enough?

Q12 · Page 126 A pack contains 56 tablets. Regimen: one tablet four times a day for 4 weeks. How many packs?

Packs = ?

Q13 · Page 126 A pack of 84 tablets is designed to last 1 month. If the patient takes one dose twice a day, what is the dose?

Dose = ?

5.2 Oral Liquid Formulations

Lecture Problems

Slide 9 A prescription of indigestion mixture with dosage regimen of 20 ml four times daily for 5 days.

A. How many 5 ml spoonful of mixture required for each dose?

B. Calculate the number of doses.

C. Calculate the total volume of mixture that should be supplied.

Textbook Problems

Example 7.3 · Page 115 Indigestion mixture: 15 mL four times a day for 5 days. How many 5-mL spoonfuls per dose? Number of doses and total volume?

Spoonfuls, doses, volume = ?

Example 7.4 · Page 115 A 100-mL bottle of antihistamine mixture; dosage two 5-mL spoonfuls four times a day. How many days will it last?

Days = ?

Q29 · Page 128 Drug X: 240 mg four times a day; send 100 mL of 120 mg/5 mL mixture. How many days will it last?

Days = ?

Example 7.8 · Page 119 360 mg of paracetamol four times a day for 3 days, supplied as 120 mg/5 mL suspension. Volume to supply?

Volume = ?

Q28 · Page 127 Drug Z: 62.5 mg three times a day for 5 days; available as 250 mg/5 mL suspension. Volume of each dose and total?

Dose volume and total = ?

Q31 · Page 128 Elixir 150 mg/5 mL. Prescription: 225 mg four times a day for 7 days. Volume to supply?

Volume = ?

Q14 · Page 126 How many 5-mL doses are contained in 120 mL of mixture?

Doses = ?

Q17 · Page 126 How many 15-mL doses are contained in 240 mL of mixture?

Doses = ?

Q15 · Page 126 30 mL of mouthwash twice a day for a 14-day course. Volume to supply?

Volume = ?

Q16 · Page 126 Indigestion mixture: 15 mL four times a day for 2/52. How much to dispense?

Volume = ?

Q18 · Page 126 10 mL of laxative solution at night. Volume used in a month?

Volume = ?

Q21 · Page 127 7.5 mL of a mixture four times a day for 3/52. Volume to supply?

Volume = ?

Q19 · Page 126 A 150-mL bottle of cough elixir. Each of two children gets 5 mL four times a day for 3 days. How many doses left?

Doses left = ?

5.3 Dose Volumes Less Than 5 ml

Lecture Problems

Slide 12 A prescription states: 2.5 ml three times daily for 4/7. What volume of drug should be dispensed?

Volume = ?

Textbook Problems

Q20 · Page 126 A prescription for a young child reads: 2.5 mL three times a day for 4/7. What volume should be dispensed? (Same as the lecture question on Slide 12.)

Volume = ?

Example 7.5 · Page 117 A very young child is prescribed 2.0 mL of an analgesic suspension three times a day for 4/7. How many mL to dispense?

Volume = ?

Example 7.9 · Page 120 What volume of paracetamol suspension (250 mg/5 mL) would provide a dose of 200 mg?

Volume = ?

Q22 · Page 127 A pack contains 30 mL of a syrup. How many days will it last if the dose is 0.5 mL three times a day?

Days = ?

5.4 Injections & Units

Textbook Problems

Example 7.10 · Page 120 A patient requires an IM injection of 0.5 mg of adrenaline. Adrenaline injection contains 100 micrograms/mL. How many mL?

Volume = ?

Q32 · Page 128 15 mg of morphine sulfate SC every 4 hours for 16 hours; injection 20 mg/mL. Volume per dose and total morphine?

Volume per dose and total = ?

Example 7.11 · Page 122 Nystatin: 500 000 units every 6 hours. Suspension 100 000 units/mL. Total daily dose and volume for 5 days?

Daily dose and volume = ?

Q30 · Page 128 A solution contains 500 units/mL; pack size 25 mL. 2500 units three times a day for 5 days. How many packs?

Packs = ?

Example 7.12 · Page 123 Vitamin A drops contain 5000 units/mL; 40 drops/mL. How many drops give 750 units?

Drops = ?

Example 7.13 · Page 124 Product A: 8 mg vitamin A (retinyl acetate) + 400 IU ergocalciferol. Product B: 2500 IU vitamin A + 10 micrograms vitamin D. Which contains the most vitamins? (Vit A: 1 IU = 0.344 micrograms; Vit D: 40 IU = 1 microgram)

Which product?

5.5 Doses Based on Body Weight

Lecture Problems

Slide 15 The usual dose of chlorambucil is 150 mcg/kg once a day. How many milligrams should be administered to a person weighing 154 lb?

Dose (mg) = ?

Slide 16 Normal gentamicin dose is 3 mg/kg/day in three divided doses. What would be the single (8-hour) dose for a patient weighing 165 lb scheduled to receive only 40% of the usual dose (renal impairment)?

Single dose = ?

Slide 17 A patient weighing 120 lb was administered 2.1 g of a drug supposed to be dosed at 30 mg/kg. Correct, overdose or underdose?

Verdict?

5.6 Doses Based on Body Surface Area (BSA)

Lecture Problems

Slides 18-20 BSA = √(Ht (cm) × Wt (kg) / 3600) (average adult = 1.73 m^2). If the dose of a drug is 5 mg/m^2, what would be the dose for a patient measuring 165 cm and weighing 65 kg?

Dose = ?

Slide 21 The chemotherapy regimen "CAF" during a 21-day cycle is: cyclophosphamide 500 mg/m^2, D-1; doxorubicin 50 mg/m^2, D-1; 5-fluorouracil 500 mg/m^2, D-1, 8. Calculate the dose of each drug per cycle for a patient with a BSA of 1.9 m^2.

Doses per cycle = ?

Slide 22 VAD regimen (multiple myeloma), 28-day cycle: vincristine 0.4 mg/d CIVI, D 1–4; doxorubicin 9 mg/m²/d CIVI, D 1–4; dexamethasone 40 mg/d PO, D 1–4, 9–12, 17–20. Total of each drug per cycle for BSA 1.65 m²?

Totals = ?

Answers · Lecture 5

5.1 Solid Dose Units (Tablets, Capsules…)

Slide 8 **10 capsules** — Number of capsules: 2 (per day)
Duration is 5/7 = 5 days
Total number of capsules: 2 × 5 = 10

Example 7.1 · Page 113 **42 tablets** — Per day: 2 × 3 = 6
7 days: 6 × 7 = 42

Example 7.2 · Page 114 **6** — 3/7 = 3 days
2 × 3 = 6

Q1 · Page 125 **672** — 2 × 4 = 8/day
3 months = 3 × 28 = 84 days
8 × 84 = 672

Q7 · Page 125 **168** — 2 × 3 = 6/day
1/12 = 28 days → 6 × 28 = 168

Example 7.6 · Page 117 **168 tablets** — 20 mg = two 10-mg tablets
Every 8 hours = 3 times a day → 6 tablets/day
4 weeks = 28 days → 6 × 28 = 168

Q23 · Page 127 **15** — 50 mg = ½ tablet
½ × 3 × 10 = 15

Q24 · Page 127 **224** — 125 mg = 2 capsules
2 × 4 × 28 = 224

Q25 · Page 127 **a. 4 tablets** — 2 mg = 2000 micrograms / 500 = 4
b. 3 tablets — 1500 / 500 = 3
c. 63 — 4 × 4 × 3 = 48
3 × 1 × 5 = 15 (1.5 mg once a day)
48 + 15 = 63

Q26 · Page 127 **20** — 500 mg = 2 capsules
2 × 2 × 5 = 20

Example 7.7 · Page 118 **21 × 4-mg and 21 × 1-mg** — Week 1: 8 mg = 2 × 4 mg → 14 × 4-mg
Week 2: 7 mg = 1 × 4 mg + 3 × 1 mg → 7 × 4-mg + 21 × 1-mg
Total: 21 × 4-mg, 21 × 1-mg

Q27 · Page 127 **33 × 5-mg; 36 × 2.5-mg** —

stage	5-mg	2.5-mg
7.5 mg × 3 × 5	15	15
5 mg × 4 × 3	12	0
5 mg × 2 × 3	6	0
2.5 mg × 3 × 7	0	21
Total	33	36

7.5 mg = one 5-mg + one 2.5-mg

Q2 · Page 125 **33 days** — 100 / 3 = 33.3 → 33 days

Q3 · Page 125 **33 days** — 2 × 3 = 6 doses/day
200 / 6 = 33.3 → 33 days

Q6 · Page 125 **14 days** — 35 × 2 = 70/night
1000 / 70 = 14.28 → 14 days

Q4 · Page 125 **39** — 6 × 5 = 30
3 × 3 = 9
30 + 9 = 39

Q8 · Page 125 **58** — 3 × 2 × 3 = 18
2 × 2 × 3 = 12
1 × 2 × 14 = 28
18 + 12 + 28 = 58

Q5 · Page 125 **8 of A; 112 of B** — A: 1 × 8 = 8
B: 2 × 56 days = 112

Q11 · Page 126 **28** — 3 months = 84 days
84 / 3 = 28

Q9 · Page 126 **112** — 2 eyes × 2 times = 4/day
4 × 28 = 112

Q10 · Page 126 **Yes** — 2 × 4 = 8/day × 21 days = 168 → exactly enough

---

Q12 · Page 126 **2** — 4 × 28 = 112 tablets
112 / 56 = 2 packs

Q13 · Page 126 **1.5 tablets** — 2 doses × 28 days = 56 doses
84 / 56 = 1.5 tablets per dose

5.2 Oral Liquid Formulations

Slide 9

A. 4 spoonfuls — 20 / 5 = 4 × 5-ml spoonfuls

B. 20 doses — 4 × 5 = 20 doses

C. 400 ml — 20 doses × 20 ml = 400 ml

Example 7.3 · Page 115

3 spoonfuls; 20 doses; 300 mL — 15 / 5 = 3 spoonfuls
4 × 5 = 20 doses
20 × 15 = 300 mL

Example 7.4 · Page 115

2.5 days — 10 mL × 4 = 40 mL/day
100 / 40 = 2.5 days

Q29 · Page 128

2.5 days — 240 mg = 10 mL
10 × 4 = 40 mL/day
100 / 40 = 2.5 days

Example 7.8 · Page 119

180 mL —

paracetamol (mg)	360	120
suspension (mL)	y	5

y = 5 × 360 / 120 = 15 mL
15 × 4 × 3 = 180 mL

Q28 · Page 127

1.25 mL per dose; 18.75 mL — 62.5 × 5 / 250 = 1.25 mL
1.25 × 3 × 5 = 18.75 mL

Q31 · Page 128

210 mL — 225 × 5 / 150 = 7.5 mL
7.5 × 4 × 7 = 210 mL

Q14 · Page 126

24 — 120 / 5 = 24

Q17 · Page 126

16 — 240 / 15 = 16

Q15 · Page 126

840 mL — 30 × 2 × 14 = 840 mL

Q16 · Page 126

840 mL — 15 × 4 = 60 mL/day
2/52 = 14 days → 60 × 14 = 840 mL

Q18 · Page 126

280 mL — 10 × 28 = 280 mL

Q21 · Page 127

630 mL — 7.5 × 4 = 30 mL/day × 21 days = 630 mL

Q19 · Page 126

6 — 150 / 5 = 30 doses
Used: 2 × 4 × 3 = 24
30 − 24 = 6

5.3 Dose Volumes Less Than 5 ml

Slide 12 **30 ml** — Number of doses: 3
Volume of each dose: 2.5 ml
Daily: 3 × 2.5 = 7.5 ml
Total for 4 days: 7.5 × 4 = 30 ml

Q20 · Page 126 **30 mL** — 2.5 × 3 × 4 = 30 mL

Example 7.5 · Page 117 **24 mL** — 2.0 × 3 = 6.0 mL/day
4 × 6 = 24 mL

Example 7.9 · Page 120 **4 mL** — y = 5 × 200 / 250 = 4 mL

Q22 · Page 127 **20 days** — 0.5 × 3 = 1.5 mL/day
30 / 1.5 = 20 days

5.4 Injections & Units

Example 7.10 · Page 120 **5 mL** — 0.5 mg = 500 micrograms
y = 500 / 100 = 5 mL

Q32 · Page 128 **0.75 mL per dose; 75 mg** — 15 / 20 = 0.75 mL
Every 4 h for 16 h = 5 doses (0, 4, 8, 12, 16 h)
5 × 15 = 75 mg

Example 7.11 · Page 122 **2 000 000 units/day; 100 mL** — Every 6 h = 4 times a day → 500 000 × 4 = 2 000 000 units
Each dose = 5 mL → 20 mL/day × 5 = 100 mL

Q30 · Page 128 **3 packs** — 2500 / 500 = 5 mL per dose
5 × 3 × 5 = 75 mL
75 / 25 = 3 packs

Example 7.12 · Page 123 **6 drops** — x = 40 × 750 / 5000 = 6

Example 7.13 · Page 124 **Product A** — A: 8000 / 0.344 = 23 256 IU vitamin A
A: 400 / 40 = 10 micrograms vitamin D
B: 2500 IU vitamin A, 10 micrograms vitamin D
Same vitamin D, more vitamin A in A

5.5 Doses Based on Body Weight

Slide 15 **10.5 mg** — 154 lb / 2.2 = 70 kg
150 mcg × 70 = 10 500 mcg = 10.5 mg

Slide 16 **30 mg** — 165 / 2.2 = 75 kg
3 × 75 = 225 mg/day
40% → 225 × 0.4 = 90 mg/day
÷ 3 doses = 30 mg every 8 hours

Slide 17 **Overdose** — 120 / 2.2 = 54.5 kg
30 × 54.5 = 1636 mg ≈ 1.64 g
2.1 g > 1.64 g → overdose

5.6 Doses Based on Body Surface Area (BSA)

Slides 18-20 **8.6 mg** — BSA = √(165 × 65 / 3600) = √2.979 = 1.73 m²
Dose = 5 × 1.73 = 8.6 mg

Slide 21 **Cyclophosphamide 950 mg; doxorubicin 95 mg; 5-FU 1900 mg** —

drug	per dose	days	per cycle
Cyclophosphamide	500 × 1.9 = 950 mg	1 (D-1)	950 mg
Doxorubicin	50 × 1.9 = 95 mg	1 (D-1)	95 mg
5-Fluorouracil	500 × 1.9 = 950 mg	2 (D-1, 8)	1900 mg

Slide 22 **Vincristine 1.6 mg; doxorubicin 59.4 mg; dexamethasone 480 mg** —

drug	per day	days	per cycle
Vincristine	0.4 mg	4	1.6 mg
Doxorubicin	9 × 1.65 = 14.85 mg	4	59.4 mg
Dexamethasone	40 mg	12	480 mg

6 Doses for Children

In this lecture

1. Rules Based on Age & Weight 2 lecture · 1 textbook
2. Doses Based on Body Weight 3 lecture · 9 textbook
3. Doses Based on Body Surface Area (BSA) 3 lecture · 5 textbook
4. Other Dispensing Calculations 4 lecture

6.1 Rules Based on Age & Weight

Lecture Problems

Slide 4 Drug dosage based on age — write the rule:

a. Young's rule

b. Cowling's rule

c. Fried's rule for infants

Slide 5 Drug dosage based on weight — write Clark's rule.

Clark's rule

Textbook Problems

Worked example · Page 132 BNF: "for most drugs the adult maximum dose should not be exceeded". A drug has a dose of 8 mg/kg and a maximum dose of 300 mg.

Dose for a 10-kg child and a 40-kg child = ?

6.2 Doses Based on Body Weight

Lecture Problems

Slide 9 Rx Phenytoin 30 mg/5 ml Suspension, 4 mg/kg/day for a 25-lb child.

Daily dose (mg) and volume (ml) = ?

Slide 14 The usual dose of sulfisoxazole for infants over 2 months and children is 60 to 75 mg/kg. What would be the usual range for a child weighing 44 lb?

Range = ?

Slide 16 A pediatric patient receives enalaprilat IV every 12 hours: 5 mcg/kg → 55 mcg per dose. Convert to oral enalapril 100 mcg/kg as a single daily dose. A 2.5-mg tablet is crushed and made up to 12.5 mL with sterile water. Dose in mL?

Volume = ?

Textbook Problems

Q2 · Page 138 Pethidine HCl injection 100 mg/2 mL. Child IM dose 0.5–2 mg/kg. Both extremes for a 12-year-old and the volumes.

Range and volumes = ?

Example 8.1 · Page 133 Fluconazole for mucosal candidiasis in children is 3 mg/kg daily. Calculate the dose for a child aged 3 years and a child aged 11 years and suggest a formulation (capsules 50, 150, 200 mg; suspension 50 mg/5 mL, 200 mg/5 mL).

a. 3-year-old

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b. 11-year-old

Q3 · Page 138 Griseofulvin oral dose for a child is 10 mg/kg daily; suspension 125 mg/5 mL. Dose and volume for a 5-year-old.

Dose and volume = ?

Q4 · Page 139 Using Q3, calculate the oral dose of griseofulvin and the volume of suspension for a 12-year-old child.

Dose and volume = ?

Example 8.2 · Page 135 Co-trimoxazole IV infusion: 36 mg/kg daily in two divided doses. Dose for a 5-year-old child; sterile solution 96 mg/mL.

Dose and volume = ?

Example 8.3 · Page 136 Amikacin: 15 mg/kg daily in two divided doses. Dose for a 6-month-old child; paediatric injection 50 mg/mL.

Dose and volume = ?

Q6 · Page 139 Drug X: oral dose for a 6-month-old of 8 mg/kg twice a day; suspension 40 mg/5 mL. Dose and volume each time?

Dose and volume = ?

Q1 · Page 138 Morphine sulfate for a 6-month-old child by SC injection: 200 micrograms/kg. Quantity for one dose and volume of 10 mg/mL injection?

Dose and volume = ?

Q9 · Page 139 Hydroxychloroquine sulfate oral dose for a child: 6.5 mg/kg daily. Dose for a 7-year-old?

Dose = ?

6.3 Doses Based on Body Surface Area (BSA)

Lecture Problems

Slide 6 Drug dosage on body surface area: BSA = √(Ht (cm) × Wt (kg) / 3600) (or use nomogram).

a. Child dose from the adult dose = ?

b. Child dose if the dose per m² is given = ?

Slide 8 The usual pediatric dose of ephedrine sulfate is 25 mg/m². Calculate the dose for a child weighing 18 kg and measuring 82 cm in height.

Dose = ?

Slide 15 If the usual adult dose of a drug is 100 mg, determine the approximate dose for a child with a BSA of 0.83 m².

Dose = ?

Textbook Problems

Example 8.4 · Page 137 Oral dose of methotrexate for a 5-year-old child who weighs 18 kg. Oral dose 15 mg/m² weekly.

Weekly dose = ?

Example 8.5 · Page 138 Calculate the dose of drug X for a 3-year-old boy if the dose is 120 mg/m² each day.

Dose = ?

Q7 · Page 139 The dose of drug X is 75 mg/m². Dose for a 5-year-old child of average weight?

Dose = ?

Q8 · Page 139 Using Q7, calculate the dose of drug X for a 10-year-old child of average weight.

Dose = ?

Q5 · Page 139 Didanosine 120 mg/m² every 12 hours. Dose and daily dose for a 7-year-old child who weighs 25 kg.

Dose and daily dose = ?

6.4 Other Dispensing Calculations

Lecture Problems

Slide 10 Rx Theophylline 80 mg/15 ml, give 1.5 mg as directed.

Volume = ?

Slide 11 How many drops would be prescribed in each dose of a liquid medicine if 15 mL contained 60 doses? The dispensing dropper calibrates 32 drops/ml.

Drops = ?

Slide 12 How many milliliters of a liquid medicine would provide a patient with 2 table spoonfuls twice a day for 8 days?

Volume = ?

Slide 13 It takes approximately 2 g of ointment to cover a child's leg. Ointment for total leg eczema, applied twice a day for 1 week. Which size: 15 g, 30 g, 60 g, 120 g?

Size = ?

Answers · Lecture 6

6.1 Rules Based on Age & Weight

Slide 4

a. **Dose for child = Age / (Age + 12) × Adult dose** — Age in years.

b. **Dose for child = Age at next birthday × Adult dose / 24** — Uses the age at the **next** birthday.

c. **Dose for infant = Age (in months) × Adult dose / 150** — Age in **months**.

Slide 5

Dose for child = Weight (lb) × Adult dose / 150 — 150 lb = average adult weight.

Worked example · Page 132

80 mg and 300 mg — 10 × 8 = 80 mg
40 × 8 = 320 mg → exceeds the maximum → give 300 mg

6.2 Doses Based on Body Weight

Slide 9

45.5 mg/day = 7.6 ml/day — 25 lb / 2.2 = 11.36 kg
4 × 11.36 = 45.5 mg/day
45.5 × 5 / 30 = 7.6 ml/day

Slide 14

1200 – 1500 mg — 44 / 2.2 = 20 kg
60 × 20 = 1200 mg
75 × 20 = 1500 mg

Slide 16

5.5 mL — Weight: 55 / 5 = 11 kg
Oral dose: 100 × 11 = 1100 mcg = 1.1 mg
Concentration: 2.5 mg / 12.5 mL = 0.2 mg/mL
1.1 / 0.2 = 5.5 mL

Q2 · Page 138

19.5–78 mg; 0.39–1.56 mL — 12 yrs = 39 kg
0.5 × 39 = 19.5 mg; 2 × 39 = 78 mg
50 mg/mL → 0.39 mL and 1.56 mL

Example 8.1 · Page 133

a. **42 mg = 4.2 mL of 50 mg/5 mL** — Table 8.1: 14 kg → 3 × 14 = 42 mg
42 × 5 / 50 = 4.2 mL daily

b. **≈ 105 mg (≈ 100 mg): two 50-mg capsules or 10 mL of 50 mg/5 mL** —
Estimated weight ≈ 35 kg (between 32 and 39)
3 × 35 = 105 mg

Q3 · Page 138

180 mg; 7.2 mL — 18 kg × 10 = 180 mg
180 × 5 / 125 = 7.2 mL

Q4 · Page 139

390 mg; 15.6 mL — 39 kg × 10 = 390 mg
390 × 5 / 125 = 15.6 mL

Example 8.2 · Page 135 **324 mg twice a day = 3.375 mL** — Table 8.1: 18 kg → 36 × 18 = 648 mg/day
÷ 2 = 324 mg per dose
324 / 96 = 3.375 mL

Example 8.3 · Page 136 **114 mg/day = 2.28 mL (1.14 mL twice)** — Table 8.1: 7.6 kg → 15 × 7.6 = 114 mg
114 / 50 = 2.28 mL/day → 1.14 mL per dose

Q6 · Page 139 **60.8 mg twice a day; 7.6 mL** — 7.6 kg × 8 = 60.8 mg
60.8 × 5 / 40 = 7.6 mL

Q1 · Page 138 **1.52 mg; 0.152 mL** — 7.6 kg × 200 = 1520 micrograms = 1.52 mg
1.52 / 10 = 0.152 mL

Q9 · Page 139 **149.5 mg** — 23 kg × 6.5 = 149.5 mg

6.3 Doses Based on Body Surface Area (BSA)

Slide 6 **a. BSA of child / 1.73 m² × Adult dose** — 1.73 m² = average adult BSA.
b. BSA of child × Dose per m²

Slide 8 **16 mg** — BSA = √(82 × 18 / 3600) = √0.41 = 0.64 m²
25 × 0.64 = 16 mg

Slide 15 **48 mg** — 0.83 / 1.73 × 100 = 48 mg

Example 8.4 · Page 137 **11.1 mg** — BSA (18 kg) = 0.74 m²
15 × 0.74 = 11.1 mg weekly

Example 8.5 · Page 138 **74.4 mg daily** — 3 yrs = 14 kg → BSA 0.62 m²
120 × 0.62 = 74.4 mg

Q7 · Page 139 **55.5 mg** — 5 yrs = 18 kg → BSA 0.74 m²
75 × 0.74 = 55.5 mg

Q8 · Page 139 **82.5 mg** — 10 yrs = 32 kg → BSA 1.1 m²
75 × 1.1 = 82.5 mg

Q5 · Page 139 **110.4 mg; 220.8 mg** — 25 kg → BSA 0.92 m²
120 × 0.92 = 110.4 mg
Every 12 h = 2 doses → 220.8 mg/day

6.4 Other Dispensing Calculations

Slide 10 **0.28 ml** — 80 mg → 15 ml
1.5 mg → x x = 1.5 × 15 / 80 = 0.28 ml

Slide 11 **8 drops** — 15 / 60 = 0.25 mL per dose
0.25 × 32 = 8 drops

Slide 12 **480 mL** — 1 tablespoonful = 15 mL → 2 = 30 mL
30 × 2 × 8 = 480 mL

Slide 13 **30 g** — 2 g × 2 times × 7 days = 28 g → smallest size that covers it = 30 g

7 Creatinine Clearance

In this lecture

1. Cockroft-Gault & Choosing the Weight 2 lecture
2. Ideal Body Weight (IBW) 3 lecture
3. Adjusted Body Weight (AjBW) 1 lecture
4. Creatinine Clearance & Dosing 2 lecture
5. Adjusting CrCl for Body Surface Area 1 lecture

7.1 Cockroft-Gault & Choosing the Weight

Lecture Problems

Slide 3 Cockroft-Gault equation — write it and the units.

CrCl (mL/min) = ?

Slide 4 Which weight is used in the Cockroft-Gault equation?

Weight by BMI = ?

7.2 Ideal Body Weight (IBW)

Lecture Problems

Slide 5 The ideal body weight (IBW) — write the formulas.

a. Males

---

b. Females

Slide 6 What is the IBW of a man who is 174 cm tall and weighs 70 kilograms?

IBW = ?

Slide 7 What is the IBW for a woman who is 64 inches tall and weighs 70 kilograms?

IBW = ?

7.3 Adjusted Body Weight (AjBW)

Lecture Problems

Slides 8–9 AjBW (obese patients, actual weight > IBW) = IBW + 0.4 × (actual body weight − IBW). Calculate the AjBW of a woman who is 157 cm tall and weighs 80 kilograms.

AjBW = ?

7.4 Creatinine Clearance & Dosing

Lecture Problems

Slide 10 Determine the loading and maintenance doses of gentamicin for a 76-year-old male weighing 190 lb with a height of 6 feet and serum creatinine of 2.4 mg/dL (212 mcmol/L). Loading dose 1.5 mg/kg of IBW; maintenance 1 mg/kg of IBW every 8 hours after the initial dose.

a. Loading and maintenance doses = ?

b. Extra: his creatinine clearance (Cockroft-Gault) = ?

Slide 11 What is the creatinine clearance for a 40-year-old male patient with S.Cr of 4 mg/dL (354 µmol/L)? The patient is 5 ft 4 in in height and weighs 103 kg.

CrCl = ?

7.5 Adjusting CrCl for Body Surface Area

Lecture Problems

Slide 12 Adjusted CrCl = BSA / 1.73 × eCrCl. If a patient weighing 120 lb and measuring 60 in. in height has a calculated creatinine clearance of 40 mL per minute, adjust the CrCl based on body surface area.

Adjusted CrCl = ?

Answers · Lecture 7

7.1 Cockroft-Gault & Choosing the Weight

Slide 3 **(140 − Age) × Weight × Constant / Serum creatinine** — Age in years
Weight in kg (actual, ideal or adjusted — depends on BMI)
Serum creatinine in micromol/litre
Constant = 1.23 for men; 1.04 for women

Slide 4 **<18.5 → ABW · 18.5–24.9 → IBW · ≥25 → AjBW** —

BMI	weight
< 18.5 kg/m²	Use ABW (actual)
18.5 – 24.9 kg/m²	Use IBW (ideal)
25 – 29.9 kg/m²	Use AjBW (adjusted)
≥ 30 kg/m²	Use AjBW (adjusted)

BMI = weight (kg) / height (m)²

7.2 Ideal Body Weight (IBW)

Slide 5 **a. IBW = 50 kg + 2.3 kg for each inch over 5 feet** — or in pounds: 110 lb + 5 lb for each inch over 5 feet
b. IBW = 45.5 kg + 2.3 kg for each inch over 5 feet — or in pounds: 100 lb + 5 lb for each inch over 5 feet

Slide 6 **≈ 69.6 kg** — 174 / 2.54 = 68.5 inches
Over 5 feet (60 in): 68.5 − 60 = 8.5 inches
IBW = 50 + 2.3 × 8.5 = 69.6 kg

Slide 7 **54.7 kg** — 64 − 60 = 4 inches
IBW = 45.5 + 2.3 × 4 = 54.7 kg

7.3 Adjusted Body Weight (AjBW)

Slides 8-9 **≈ 61.8 kg** — 157 / 2.54 = 61.8 inches → 1.8 inches over 5 feet
IBW = 45.5 + 2.3 × 1.8 = 49.7 kg
AjBW = 49.7 + 0.4 × (80 − 49.7) = 49.7 + 12.1 = 61.8 kg

7.4 Creatinine Clearance & Dosing

Slide 10

a. IBW 77.6 kg → loading 116.4 mg; maintenance 77.6 mg every 8 h — 6 feet = 72 inches → 12 inches over 5 feet

IBW = 50 + 2.3 × 12 = 77.6 kg

Loading = 1.5 × 77.6 = 116.4 mg

Maintenance = 1 × 77.6 = 77.6 mg every 8 hours

b. ≈ 30 mL/min — 190 lb / 2.2 = 86.4 kg; 72 in = 1.83 m → BMI = 86.4 / 1.83^2 = 25.8 → use AjBW

AjBW = 77.6 + 0.4 × (86.4 − 77.6) = 81.1 kg

CrCl = (140 − 76) × 81.1 × 1.23 / 212 = 30.1 mL/min

Slide 11

≈ 26.7 mL/min — Height = 64 in = 162.6 cm → BMI = 103 / 1.626^2 = 39 → obese → use AjBW

IBW = 50 + 2.3 × 4 = 59.2 kg

AjBW = 59.2 + 0.4 × (103 − 59.2) = 76.7 kg

CrCl = (140 − 40) × 76.7 × 1.23 / 354 = 26.7 mL/min

7.5 Adjusting CrCl for Body Surface Area

Slide 12

≈ 35 mL/min — 120 lb / 2.2 = 54.5 kg; 60 in × 2.54 = 152.4 cm

BSA = √(152.4 × 54.5 / 3600) = 1.52 m^2

Adjusted CrCl = 1.52 / 1.73 × 40 = 35.1 mL/min

8 Intravenous Therapy

In this lecture

1. Giving Sets & Drip Rate (drops/min) 2 lecture · 4 textbook
2. Converting mL/hour, mL/min & drops/min 1 lecture · 2 textbook
3. Dose Rates (mcg/kg/min) & mL/hour 2 lecture
4. Volume, Drip Rate & Duration 1 lecture · 2 textbook
5. Dose, Volume & Maximum Rate 2 lecture · 6 textbook
6. mg/hour → mL/min & drops/min 1 lecture · 2 textbook
7. Syringe Drivers (mm/hour) 4 textbook

8.1 Giving Sets & Drip Rate (drops/min)

Lecture Problems

Slide 4 Giving sets — how many drops are in 1 mL?

a. Standard giving set (SGS)

b. Microdrop giving set (burette)

Slide 5 One litre of 5% dextrose is to be given over 8 hours. What drip rate is required? (rate: 20 drops/mL)

Drip rate = ?

Textbook Problems

Example 11.1 · Page 195 One litre of 0.9% saline solution is to be given to a patient over a 6-hour period. If 20 drops = 1 mL, how many drops per minute should be administered?

Drops/min = ?

Q2 · Page 209 500 mL of 0.9% saline solution is to be given to a patient over a 6-hour period. If 20 drops = 1 mL, calculate how many drops per minute should be administered to the patient.

Drops/min = ?

Example 11.3 · Page 197 20 mL of Addiphos solution and 10 mL of multivitamin infusion are added to 500 mL of glucose 5% solution. The resultant solution is to be administered over 4 hours. The administration set is calibrated to 20 drops/mL. Calculate the number of drops per minute to be given if the patient is to receive all of the solution in the specified time.

Drops/min = ?

Q4 · Page 209 A 10-mL vial of multivitamin injection is to be added to 500 mL of glucose 5% solution. The resultant solution is to be administered to a patient over 3 hours. The administration set is calibrated at 20 drops/mL. Calculate the number of drops per minute to be given if the patient is to receive all the solution in the specified time.

Drops/min = ?

8.2 Converting mL/hour, mL/min & drops/min

Lecture Problems

Slide 6 One litre of sodium chloride 0.9% has to be given at a rate of 120 mL/hour. What is the rate in drops/min? (20 drops/mL)

Rate = ? drops/min

Textbook Problems

Q1 · Page 209 1 L of glucose 5% solution is to be given to a patient over a 5-hour period. Calculate the rate of delivery in mL/minute.

Rate = ? mL/min

Q3 · Page 209 An IV solution is to be administered at a rate of 2 mL/minute. What volume will be required if the infusion is to be maintained for 3 hours?

Volume = ?

8.3 Dose Rates (mcg/kg/min) & mL/hour

Lecture Problems

Slide 7 You have an infusion of dopamine 800 mg in 500 mL. The dose required is 2 mcg/kg/min for a patient weighing 68 kg. (20 drops/mL)

a. What is the rate in mL/hour?

b. What is the rate in drops/min?

Slide 8 An infusion pump containing 250 mg dobutamine in 50 mL is administered at a rate of 3.5 mL/hour. Convert to mcg/kg/min. (Patient weight: 70 kg)

Dose = ? mcg/kg/min

8.4 Volume, Drip Rate & Duration

Lecture Problems

Slide 9 A prescription of one litre 5% dextrose to be administered over 8 hours at a drip rate of 40 drops/min. You wish to check the drip rate. How long will it take for the infusion to finish? (20 drops/mL)

Time = ?

Textbook Problems

Example 11.2 · Page 196 The recommended flow rate for drug Y is 2–3 mL/minute at a concentration of 5 mg/mL. Drug Y is supplied in 500-mg vials for dilution in glucose 5%. Using one vial, the flow rate is 3 mL/minute and the IV set is calibrated to 20 drops/mL.

a. Volume of glucose 5% solution to use?

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b. Drops per minute?

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c. How long will the infusion take?

Q5 · Page 209 Nizatidine injection (25 mg/mL) is supplied in 4-mL vials. The BNF states that for continuous infusion "dilute 300 mg in 150 mL and give at a rate of 10 mg/hour". If this procedure is adopted, calculate:

a. How many vials are required?

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b. The flow rate (mL/hour)?

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c. The length of time the infusion will take?

8.5 Dose, Volume & Maximum Rate

Lecture Problems

Slide 10 A 73 kg patient requires drug X by IV infusion. The dose is 400 mcg/kg. The final concentration of solution must not exceed 150 mcg/mL. The drug is available as a vial of 120 mg. The infusion needs to be delivered in 3.5 hours.

1. Calculate the dose.

2. Calculate the volume required by the patient.

3. What is the rate in mL/min?

4. What is the total volume of the solution prepared (whole vial)?

Slide 11 The required dose of metoprolol is 75 mg at a rate not exceeding 8 mg/min. Metoprolol injection is available as 20 mg/mL.

a. Volume of metoprolol injection required?

b. Infusion rate (mL/min)?

c. Infusion duration?

Textbook Problems

Example 11.4 · Page 198 A 56-kg female patient requires amphotericin by IV infusion at a dose of 250 micrograms/kg. The concentration of the final solution must not be more than 100 micrograms/mL. A vial of amphotericin contains 50 mg. The solution has to be delivered in 2.5 hours.

a. Dose = ?

b. Volume of IV solution (maximum concentration)?

c. Rate in mL/minute?

d. Total volume if a 50-mg vial is used?

Example 11.6 · Page 201 The required dose of furosemide by slow IV infusion is 50 mg at a rate not exceeding 4 mg/minute. Furosemide injection contains 10 mg in 1 mL. Calculate the volume of injection required and the infusion rate (mL/minute).

a. Volume of injection?

b. Time and infusion rate?

Q8 · Page 210 The recommended dose of drug X is 75 mg by slow infusion at a rate not exceeding 0.3 g/hour. Drug X is available as an injection containing 15 mg in 1 mL. Calculate the volume of injection required and the infusion rate in mL/minute.

Volume and rate = ?

Example 11.5 · Page 200 Phenytoin: 18 mg/kg, rate not exceeding 50 mg/minute. Patient: woman, 56 kg. Injection: 5-mL ampoules, 50 mg/mL. The prescriber would like an infusion volume of 100 mL and a dose rate of 25 mg/minute. Is the request possible and, if so, what is the flow rate per minute?

a. Dose and volume of injection?

b. Possible? Flow rate?

Q6 · Page 209 A pharmacist prepares a solution of vancomycin containing 1 g in 200 mL. The rate of infusion should not exceed 10 mg/minute.

a. Maximum flow rate (mL/min)?

b. Flow rate if given over 2 hours?

Q7 · Page 209 A baby weighing 3.4 kg requires phenytoin at a dose of 20 mg/kg. A 5-mL vial of phenytoin injection containing 50 mg/mL is available. A suitable volume of injection is diluted up to 50 mL with sodium chloride 0.9%.

a. Volume of phenytoin injection to use?

b. Maximum flow rate if the rate must not exceed 3 mg/kg/minute?

c. Minimum flow rate if completed in 1 hour?

8.6 mg/hour → mL/min & drops/min

Lecture Problems

Slide 12 A drug is available as an infusion solution at a concentration of 0.25 mg/mL. The prescribed flow rate is 5 mg/hour. Calculate the flow rate in mL/min and drops/min (1 mL = 20 drops).

Flow rate = ?

Textbook Problems

Example 11.7 · Page 203 A drug is provided as an infusion solution at a concentration of 0.15 mg/mL. The prescribed flow rate is 3 mg/hour. Calculate the flow rate in mL/minute and drops/minute if 1 mL = 20 drops.

Flow rate = ?

Q9 · Page 210 A drug is provided as an infusion solution at a concentration of 2 micrograms/mL. The prescribed flow rate is 0.05 mg/hour. Calculate the flow rate in mL/minute and drops/minute if 1 mL = 25 drops.

Flow rate = ?

8.7 Syringe Drivers (mm/hour)

Textbook Problems

Example 11.8 · Page 204 Apomorphine injection (10 mg/mL) is diluted with an equal volume of sodium chloride 0.9% and put in a 10-mL syringe that is 60 mm long. The patient requires 2.5 mg of apomorphine per hour. At what rate (mm/hour) should the syringe driver be set?

Rate = ? mm/hour

Example 11.9 · Page 205 A 100-mg dose of an analgesic has to be infused over 5 hours. The drug is in 1-mL vials each containing 50 mg. The syringe driver holds 20 mL and delivers 2 mL/hour. How many vials and what volume of diluent are needed to fill the 20-mL syringe?

Vials and diluent = ?

Q10 · Page 210 A syringe driver is designed to contain 5 mL of injection and has a length of 60 mm. Injection Y contains 4 mg/mL. If a patient requires 2.5 mg of drug Y per hour, at what rate setting in mm/hour should the syringe driver be fixed?

Rate = ? mm/hour

Q11 · Page 210 A syringe driver is set at 5 mm/hour. The length of the syringe driver is 80 mm and it holds 10 mL of injection solution. The injection contains 5 micrograms of drug X in 1 mL. What dose of drug X will the syringe driver deliver in 3 hours?

Dose = ?

Answers · Lecture 8

8.1 Giving Sets & Drip Rate (drops/min)

Slide 4

a. 20 drops/mL for crystalloids · 15 drops/mL for colloids — Crystalloids: e.g. sodium chloride, dextrose.

Colloids: e.g. blood.

b. 60 drops/mL

Slide 5

42 drops/min — Step 1: 1000 mL × 20 = 20 000 drops

Step 2: 8 h × 60 = 480 min

Step 3: 20 000 / 480 = 41.67 → round up → 42 drops/min

Example 11.1 · Page 195

56 drops/min — 6 h = 360 min

1000 mL × 20 = 20 000 drops

20 000 / 360 = 55.5 → 56 drops/min

Q2 · Page 209

28 drops/min — 500 × 20 = 10 000 drops

6 × 60 = 360 min

10 000 / 360 = 27.8 → 28 drops/min

Example 11.3 · Page 197

44 drops/min — Total volume = 20 + 10 + 500 = 530 mL

530 × 20 = 10 600 drops

4 × 60 = 240 min

10 600 / 240 = 44.17 → 44 drops/min

Q4 · Page 209

≈ 57 drops/min (56.7) — Total volume = 10 + 500 = 510 mL

510 × 20 = 10 200 drops

3 × 60 = 180 min

10 200 / 180 = 56.7 → 57 drops/min

8.2 Converting mL/hour, mL/min & drops/min

Slide 6

40 drops/min — Step 1: 120 mL/hour = 120 mL / 60 min

Step 2: 120 / 60 = 2 mL/min

Step 3: 2 mL × 20 drops = 40 drops/min

Q1 · Page 209

3.33 mL/min — 5 h = 300 min

1000 / 300 = 3.33 mL/min

Q3 · Page 209

360 mL — 3 h = 180 min

2 mL/min × 180 min = 360 mL

8.3 Dose Rates (mcg/kg/min) & mL/hour

Slide 7

a. ≈ 5.1 mL/hour — 2 × 68 = 136 mcg/min
136 × 60 = 8160 mcg/hour = 8.16 mg/hour
800 mg → 500 mL
8.16 mg → x
x = 8.16 × 500 / 800 = 5.1 mL/hour (slide: ≈ 5 mL/hour)

b. ≈ 1.7 drops/min — 5.1 mL/hour × 20 = 102 drops/hour
102 / 60 = 1.7 drops/min (slide: 100 / 60 = 1.7)

Slide 8

≈ 4.2 mcg/kg/min — 250 mg → 50 mL
x → 3.5 mL → x = 17.5 mg/hour
17.5 / 60 = 0.2917 mg/min = 291.7 mcg/min
291.7 / 70 = 4.17 ≈ 4.2 mcg/kg/min

8.4 Volume, Drip Rate & Duration

Slide 9

500 min = 8 h 20 min — 1000 mL × 20 = 20 000 drops
20 000 / 40 = 500 min
500 / 60 = 8.33 hours = 8 h 20 min

Example 11.2 · Page 196

a. 100 mL — 5 mg → 1 mL
500 mg → y → y = 100 mL

b. 60 drops/min — 3 mL/min × 20 = 60 drops/min

c. 33.3 min — 100 mL / 3 mL per min = 33.3 min

Q5 · Page 209

a. 3 vials — One vial = 25 × 4 = 100 mg
300 / 100 = 3 vials

b. 5 mL/hour — Concentration = 300 mg / 150 mL = 2 mg/mL
10 mg/hour ÷ 2 mg/mL = 5 mL/hour

c. 30 hours — 150 mL / 5 mL per hour = 30 h (or 300 mg / 10 mg per hour)

8.5 Dose, Volume & Maximum Rate

Slide 10

1. **29 200 mcg = 29.2 mg** — 400 mcg × 73 kg = 29 200 mcg
2. **≈ 194.7 mL** — Maximum concentration 150 mcg/mL:
   29 200 / 150 = 194.7 mL
3. **≈ 0.93 mL/min** — 3.5 h = 210 min
   194.7 / 210 = 0.927 mL/min
4. **800 mL** — 120 mg = 120 000 mcg
   120 000 / 150 = 800 mL

Slide 11

a. 3.75 mL — 75 / 20 = 3.75 mL

b. 0.4 mL/min — 8 mg/min ÷ 20 mg/mL = 0.4 mL/min

c. ≈ 9.4 min — 75 mg / 8 mg per min = 9.375 min (or 3.75 mL / 0.4 = 9.375 min)

Example 11.4 · Page 198

a. 14 000 micrograms — 250 × 56 = 14 000 micrograms

b. 140 mL — 14 000 / 100 = 140 mL

c. 0.93 mL/min — 2.5 h = 150 min

140 / 150 = 0.93 mL/min

d. 500 mL — 50 mg = 50 000 micrograms

50 000 / 100 = 500 mL (only 140 mL is given)

Example 11.6 · Page 201

a. 5 mL — 50 / 10 = 5 mL

b. 12.5 min · 0.4 mL/min — 50 / 4 = 12.5 min

5 mL / 12.5 min = 0.4 mL/min

Q8 · Page 210

5 mL · 0.33 mL/min — 75 / 15 = 5 mL

0.3 g/hour = 300 mg / 60 min = 5 mg/min

5 mg/min ÷ 15 mg/mL = 0.33 mL/min

Example 11.5 · Page 200

a. 900 mg · 18 mL — 18 × 56 = 900 mg

900 / 50 = 18 mL

b. Yes · 2.78 mL/min — 18 mL fits into 100 mL (made up with 0.9% saline);

25 mg/min < 50 mg/min

900 mg → 100 mL

25 mg → x = 2.78 mL/min

Q6 · Page 209

a. 2 mL/min — 1000 mg / 200 mL = 5 mg/mL

10 / 5 = 2 mL/min

b. ≈ 1.67 mL/min — 200 mL / 120 min = 1.67 mL/min

Q7 · Page 209

a. 1.36 mL — Dose = 20 × 3.4 = 68 mg

68 / 50 = 1.36 mL

b. 7.5 mL/min — 3 × 3.4 = 10.2 mg/min

Concentration = 68 mg / 50 mL = 1.36 mg/mL

10.2 / 1.36 = 7.5 mL/min

c. 0.833 mL/min — 50 mL / 60 min = 0.833 mL/min

8.6 mg/hour → mL/min & drops/min

Slide 12

0.33 mL/min · ≈ 7 drops/min — 5 mg/hour ÷ 0.25 mg/mL = 20 mL/hour

20 / 60 = 0.33 mL/min

0.33 × 20 = 6.7 → 7 drops/min

Example 11.7 · Page 203

0.33 mL/min · 7 drops/min — 3 mg/hour = 0.05 mg/min

0.05 / 0.15 = 0.33 mL/min

0.33 × 20 = 6.7 → 7 drops/min

Q9 · Page 210 **0.417 mL/min · 10.4 drops/min** — 0.05 mg/hour = 50 micrograms/hour
50 / 2 = 25 mL/hour
25 / 60 = 0.417 mL/min
0.417 × 25 = 10.4 drops/min (≈ 10)

8.7 Syringe Drivers (mm/hour)

Example 11.8 · Page 204 **3 mm/hour** — 5 mL injection + 5 mL NaCl → 50 mg in 10 mL (60 mm)
50 mg → 60 mm
2.5 mg → x = 3 mm
So 3 mm/hour

Example 11.9 · Page 205 **4 vials · 16 mL diluent** — 100 / 5 = 20 mg/hour → 2 mL must contain 20 mg
20 mL → 200 mg
200 / 50 = 4 vials (4 mL)
Diluent = 20 − 4 = 16 mL

Q10 · Page 210 **7.5 mm/hour** — 5 mL × 4 = 20 mg in 60 mm
2.5 mg → 2.5 × 60 / 20 = 7.5 mm
So 7.5 mm/hour

Q11 · Page 210 **9.375 micrograms** — 10 mL × 5 = 50 micrograms in 80 mm
3 h × 5 mm = 15 mm
15 × 50 / 80 = 9.375 micrograms